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**Methods for determination of nitrogen
oxides in flue gas**

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Contents

	Page
Introduction	1
1 Scope	1
2 Normative references	1
3 Terms and definitions	3
4 General matters	3
5 Type and summary of analysis methods	3
6 Method of gas sampling	5
6.1 Evacuated flask method	5
6.2 Syringe method	18
6.3 Absorption bottle method	22
7 Chemical analysis method	24
7.1 Zinc reduction naphthylethylenediamine absorptiometry (Zn-NEDA method)	24
7.2 Naphthylethylenediamine absorptiometry (NEDA method)	27
7.3 Ion chromatography	30
7.4 Phenol disulfonic acid absorptiometry (PDS method)	36
7.5 Saltzman absorptiometry	38
8 Automated measuring method	40
9 Recording of analysis results	40
9.1 Items to be recorded	40
9.2 Method to obtain analysis value	40
Annex JA (normative) Simultaneous analysis of nitrogen oxides, sulfur ox- ides and hydrogen chloride by ion chromatography	41
Annex JB (informative) Comparison table between JIS and corresponding International Standard	45

Foreword

This translation has been made based on the original Japanese Industrial Standard revised by the Minister of Economy, Trade and Industry through deliberations at the Japanese Industrial Standards Association Committee as the result of proposal for revision of Japanese Industrial Standard submitted by Japan Environmental Measurement and Chemical Analysis Association (JEMCA)/Japanese Standards Association (JSA) with the draft being attached, based on the provision of Article 12 Clause 1 of the Industrial Standardization Law applicable to the case of revision by the provision of Article 14.

Consequently **JIS K 0104: 2000** is replaced with this Standard.

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Methods for determination of nitrogen oxides in flue gas

Introduction

This Japanese Industrial Standard has been prepared based on the first edition of ISO 11564 published in 1998 with some modifications of the technical contents by addition of the matters not specified in the corresponding International Standard and deletion of parts of the corresponding matters.

The portions with continuous sidelines or dotted underlines in this Standard are the matters in which the contents of the corresponding International Standard have been modified. A list of modifications with the explanations is given in Annex JB. Annex JA specifies the matters not in the corresponding International Standard.

1 Scope

This Standard specifies the methods to determine nitrogen oxides contained in flue gas.

The flue gasses stated in this Standard are defined as the gases which are exhausted into a flue, chimney, duct, etc. from such processes as fuel combustion, metal surface treatment, inorganic and organic chemical reaction and denitration process.

NOTE: The International Standard corresponding to this Standard and the symbol of degree of correspondence are as follows.

ISO 11564:1998 *Stationary source emissions—Determination of the mass concentration of nitrogen oxides—Naphthylethylenediamine photometric method (MOD)*

In addition, symbols which denote the degree of correspondence in the contents between the relevant International Standard and JIS are IDT (identical), MOD (modified), and NEQ (not equivalent) according to ISO/IEC Guide 21-1.

2 Normative references

The following standards contain provisions which, through reference in this text, constitute provisions of this Standard. The most recent editions of the standards (including amendments) indicated below shall be applied.

JIS B 7982	<i>Automated measuring systems and analyzers for nitrogen oxides in flue gas</i>
JIS K 0050	<i>General rules for chemical analysis</i>
JIS K 0055	<i>General rules for calibration method of gas analyzer</i>
JIS K 0095	<i>Methods for sampling of flue gas</i>
JIS K 0115	<i>General rules for molecular absorptiometric analysis</i>

JIS K 0127	<i>General rules for ion chromatographic analysis</i>
JIS K 0211	<i>Technical terms for analytical chemistry (General part)</i>
JIS K 0212	<i>Technical terms for analytical chemistry (optical part)</i>
JIS K 0213	<i>Technical terms for analytical chemistry (Electrochemistry part)</i>
JIS K 0214	<i>Technical terms for analytical chemistry (Chromatography part)</i>
JIS K 0215	<i>Technical terms for analytical chemistry (Analytical instrument part)</i>
JIS K 0216	<i>Technical terms for analytical chemistry (Environmental part)</i>
JIS K 0557	<i>Water used for industrial water and wastewater analysis</i>
JIS K 1101	<i>Oxygen</i>
JIS K 1107	<i>Nitrogen</i>
JIS K 8013	<i>Zinc powder (Reagent)</i>
JIS K 8019	<i>Sodium nitrite (Reagent)</i>
JIS K 8032	<i>Acetonitrile (Reagent)</i>
JIS K 8085	<i>Ammonia solution (Reagent)</i>
JIS K 8180	<i>Hydrochloric acid (Reagent)</i>
JIS K 8197	<i>N-1-Naphthylethylenediamine dihydrochloride (Reagent)</i>
JIS K 8230	<i>Hydrogen peroxide (Reagent)</i>
JIS K 8267	<i>Sodium formate (Reagent)</i>
JIS K 8295	<i>Glycerol (Reagent)</i>
JIS K 8355	<i>Acetic acid (Reagent)</i>
JIS K 8371	<i>Sodium acetate trihydrate (Reagent)</i>
JIS K 8548	<i>Potassium nitrate (Reagent)</i>
JIS K 8574	<i>Potassium hydroxide (Reagent)</i>
JIS K 8576	<i>Sodium hydroxide (Reagent)</i>
JIS K 8586	<i>Sulfanilic acid (Reagent)</i>
JIS K 8622	<i>Sodium hydrogen carbonate (Reagent)</i>
JIS K 8625	<i>Sodium carbonate (Reagent)</i>
JIS K 8741	<i>Sulfuric acid, fuming (Reagent)</i>
JIS K 8798	<i>Phenol (Reagent)</i>
JIS K 8863	<i>Boric acid (Reagent)</i>
JIS K 8866	<i>Sodium tetraborate decahydrate (Reagent)</i>
JIS K 8951	<i>Sulfuric acid (Reagent)</i>

JIS K 8983	<i>Copper (II) sulfate pentahydrate (Reagent)</i>
JIS K 9066	<i>Sulfanilamide (Reagent)</i>
JIS K 9704	<i>2-Amino-2-hydroxymethyl-1,3-propanediol (Reagent)</i>
JIS K 9808	<i>2-Bis (2-hydroxyethyl)amino-2-hydroxymethyl-1,3-propanediol (Bis-Tris)</i>
JIS P 3801	<i>Filter paper (for chemical analysis)</i>
JIS T 3201	<i>Glass syringes</i>
JIS Z 8808	<i>Methods of measuring dust concentration in flue gas</i>

3 Terms and definitions

For the purposes of this Standard, the terms and definitions given in JIS K 0211, JIS K 0212, JIS K 0213, JIS K 0214, JIS K 0215 and JIS K 0216 apply.

4 General matters

The general matters shall be as follows.

- The general matters common to the chemical analysis shall be as specified in JIS K 0050.
- The general matters common to the flue gas sampling method shall be as specified in JIS K 0095.
- The general matters common to the ion chromatography shall be as specified in JIS K 0127.
- The general matters common to the molecular absorptiometric analysis shall be as specified in JIS K 0115.
- Water to be used for the analyses shall be of A2 or A3 specified in clause 4 of JIS K 0557 or the equivalent.
- Reagents with the highest grade or the appropriate quality in the relevant JIS, if any, shall be used. However, if any relevant JIS does not exist, reagents of a quality adequate for the analysis shall be used.
- Reference solution for which the preparing method is specified for respective test items, which is traceable or which is diluted to a definite concentration shall be used.
- Apparatus and implements which have the specified functions shall be used.
- Flue gas used for an analysis of nitrogen oxides, the absorbing solution of the flue gas, etc. shall not be disposed improperly.

NOTE : For traceable reagents, JCSS-marked reagents specified in the national measurement standard (Measurement Law, Article 134) are applicable.

5 Type and summary of analysis methods

The type and summary of analysis methods shall be as follows.

- Chemical analysis** The type and summary of analysis methods shall be as given in table 1.

Table 1 Type and summary of analysis methods

Type of analysis methods	Summary of analysis methods				Applicable condition
	Outline	Sampling	Determination range ^{a)} vol ppm (mg/m ³)	Target gas for analysis	
Zinc reduction naphthylethyl-enediamine absorptiometry (hereafter referred to as "Zn-NEDA method".)	Make nitrogen oxides in the sample gas oxidize with ozone, and absorb into the absorption solution to be converted to nitrate ion. After making this nitrate ion be reduced to nitrite ion with zinc powder, add sulfanilamide and naphthylethylenediamine solution for colouring and measure the absorbance (545 nm).	Evacuated flask method or syringe method Absorption solution: 0.005 mol/L sulfuric acid Quantity of solution: 20 ml	Evacuated flask method: 1 to 50 (2 to 100) Syringe method: 5 to 250 (10 to 510)	NO + NO ₂	
Naphthyl-ethylenediamine absorptiometry (hereafter referred to as "NEDA method".)	Make nitrogen oxides in the sample gas absorb into the alkaline absorption solution to be converted to nitrite ion. Add sulfanilamide and naphthylethylenediamine dihydrochloride solution for colouring and measure the absorbance (545 nm).	Evacuated flask method or syringe method Absorption solution: alkaline hydrogen peroxide solution- sodium formate solution Quantity of solution: 50 ml	Evacuated flask method: 3 to 500 (5 to 1 000) Syringe method: 7 to 1 200 (13 to 2 500)	NO + NO ₂	
Ion chromatography	Make nitrogen oxides in the sample gas oxidize with ozone or oxygen, absorb into the absorption solution to be converted to nitrate ion. Introduce it into an ion chromatograph, and record its chromatogram.	Evacuated flask method or syringe method Absorption solution: 0.005 mol/L sulfuric acid-hydrogen peroxide solution (1+99) Quantity of solution: 20 ml	Evacuated flask method: 4 to 1 400 (8 to 2 800) Syringe method: 19 to 7 200 (39 to 14 000)	NO + NO ₂	
Phenol disulfonic acid absorptiometry (hereafter referred to as "PDS method".)	Make nitrogen oxides in the sample gas oxidize with ozone or oxygen, and absorb into the absorption solution to be converted to nitrate ion. Add phenol disulfonic acid for colouring and measure the absorbance (400 nm).	Evacuated flask method or syringe method Absorption solution: 0.005 mol/L sulfuric acid-hydrogen peroxide solution (1+99) Quantity of solution: 20 ml	Evacuated flask method: 10 to 300 (20 to 620) Syringe method: 12 to 4 200 (24 to 8 400)	NO + NO ₂	See 7.4.1.
Saltzman absorptiometry	Pass nitrogen dioxide in the sample gas through the absorption colouring liquid for colouring and measure the absorbance (545 nm).	Absorption bottle method Absorption colouring liquid: sulfanilic acid-naphthylethyl-enediamine acetic acid solution Quantity of solution: 25 ml	5 to 200 (10 to 400)	NO ₂	See 7.5.1.
Note ^{a)} Indicates the following volume of the sample gas when sampling: about 1 000 ml in case of evacuated flask (1 L). However, 500 ml in case of NEDA method, 200 ml in case of the syringe (200 ml), 100 ml in case of absorption bottle. In case of NEDA method, when the concentration exceeds 1 000 mg/m ³ , the sample solution for analysis can be measured up to 5 000 mg/m ³ by diluting or taking partly.					

In addition to the methods as specified in table 1, the simultaneous analysis of nitrogen oxides, sulfur oxides and hydrogen chloride by ion chromatography is specified in Annex JA.

- b) **Automated measuring method** The automated measuring method for nitrogen oxides in flue gas shall be in accordance with the method specified in JIS B 7982.

6 Method of gas sampling

The sampling method of sample gas to be used for chemical analysis shall be in accordance with the evacuated flask method, the syringe method and the absorption bottle method.

Choose the spot where representative sample gas for analysis can be taken, sample principally at least twice the sample gas at shorter intervals at the same sampling spot, and use all of them for analysis.

6.1 Evacuated flask method

The evacuated flask method shall be as follows.

This method shall be applied to Zn-NEDA method, NEDA method, ion chromatography and PDS method.

6.1.1 Preparation of reagents and reagent solution

a) Reagents

- 1) Sulfuric acid Specified in JIS K 8951.
- 2) Copper (II) sulfate pentahydrate Specified in JIS K 8983.
- 3) Sodium hydroxide Specified in JIS K 8576.
- 4) Sodium carbonate Specified in JIS K 8625.
- 5) Sodium formate Specified in JIS K 8267.
- 6) Hydrogen peroxide Specified in JIS K 8230.
- 7) Oxygen Specified in JIS K 1101.

b) Preparation of reagent solution

- 1) Sulfuric acid (1+17)
- 2) Copper (II) sulfate solution Dissolve 1 g of copper (II) sulfate pentahydrate in water, transfer it into a 1 000 ml one-mark volumetric flask and add water up to the marked line. Take 10 ml of this solution into a 100 ml one-mark volumetric flask and add water up to the marked line.
- 3) Sodium hydroxide solution (40 g/L) For filling the washing bottle. Dissolve 4 g of sodium hydroxide in water to make 100 ml.
- 4) Absorption solution Use any of the following solution.
 - 4.1) In case of zinc reduction naphthylethylenediamine absorptiometry (Zn-NEDA method)

0.005 mol/l sulfuric acid Take about 800 ml of water into a 1 000 ml one-mark volumetric flask add 5 ml of sulfuric acid (1+17) and then add water up to the marked line.

4.2) In case of naphthylethylenediamine method (NEDA method)

Alkaline hydrogen peroxide solution- sodium formate solution Dissolve 48 g of sodium hydroxide, 40 g of sodium carbonate and 5 mg of sodium formate in about 800 ml of water, add 20 ml of hydrogen peroxide solution (30 %) and then transfer it into a 1 000 ml one-mark volumetric flask while rinsing, add water up to the marked line.

4.3) In case of ion chromatography

0.005 mol/L sulfuric acid-hydrogen peroxide solution (1+99) Take about 800 ml of water into a 1 000 ml one-mark volumetric flask, add 5 ml of sulfuric acid (1+17) and 10 ml of hydrogen peroxide solution and then add water up to the marked line.

4.4) In case of phenol disulfonic acid absorptiometry (PDS method)

0.05 mol/L sulfuric acid-hydrogen peroxide solution (1+99) Take about 800 ml of water into a 1 000 ml one-mark volumetric flask, add 50 ml of sulfuric acid (1+17) and 10 ml of hydrogen peroxide solution, and then add water up to the marked line.

6.1.2 Implements and apparatus

a) Gas sampling implements

- 1) **Evacuated flask for gas sampling** As shown in figure 1, having a three-way cock, about 1 000 ml in capacity [hereafter referred to as “evacuated flask (E)”] ¹⁾. Its unobstructed capacity shall have been obtained from the mass which was weighed when filling the evacuated flask (E) with water in accordance with Annex H of JIS K 0050. The flask shown in c) of figure 1 is for the case of sampling using the evacuated flask with two cocks in NEDA method.
- 2) **Syringe for injecting absorption solution** As specified in JIS T 3201, having 20 ml or 100 ml capacity [hereafter referred to as “syringe (L)”].
- 3) **Syringe for adding oxygen or ozone** As specified in JIS T 3201, having 200 ml capacity [hereafter referred to as “syringe (M)”].

Note ¹⁾ When using the flask shown in b) of figure 1, care shall be taken not to allow the grease used to enter the flask.

b) Pressure meter

- 1) **Manometer** U-shaped open tube mercury manometer, capable of measuring pressure difference of about 27 kPa or one capable of obtaining the value equivalent to that, or closed tube mercury manometer or one capable of obtaining the value equivalent to that.
- 2) **Pressure meter** The mercury type capable of measuring 0.1 kPa to 100 kPa or one capable of obtaining the value equivalent to that. This pressure meter is used for NEDA method.

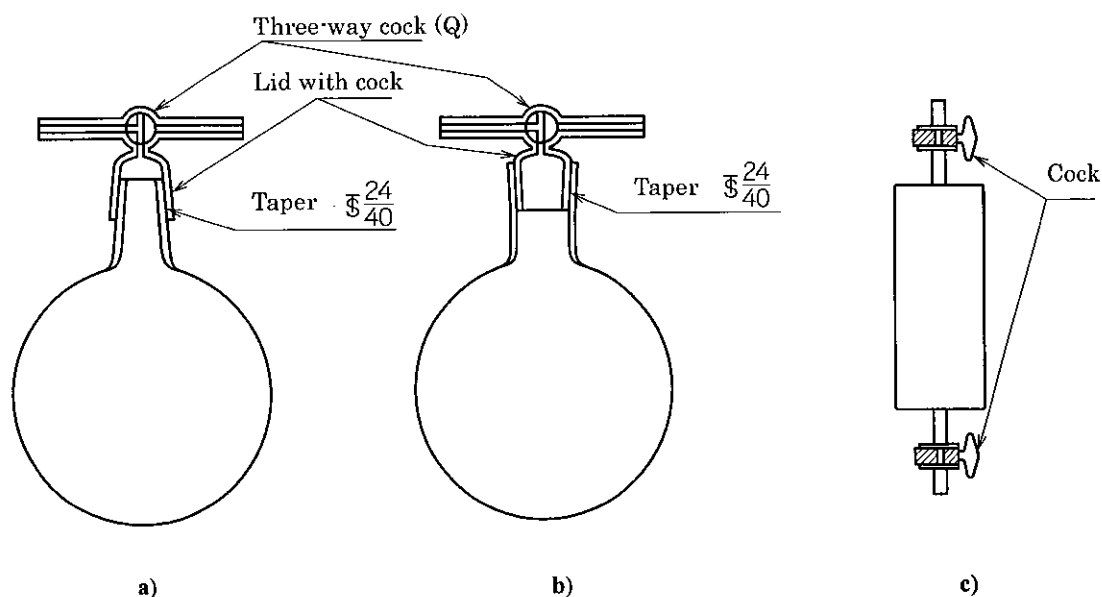


Figure 1 Example of evacuated flask for gas sampling (capacity of about 1 L)

c) **Sampling apparatus for sample gas** The apparatus, constructed as shown in figure 2, shall be equipped with the following conditions.

- 1) The sampling probe shall be made of the material, resisting corrosive gas contained in flue gas, such as glass tube quartz glass tube, or polytetrafluoroethylene tube.
- 2) In order to prevent dust from mixing into sample gas, plug suitable filter medium at end of sampling tube or at suitable place.

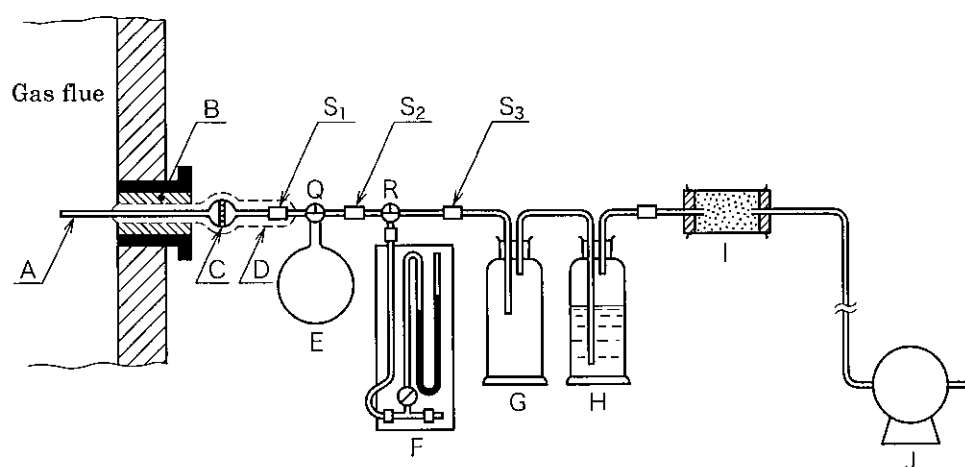
In addition, the example of filter materials when attaching them to the outside of the gas flue using the sampling apparatus shown in b) of figure 2 according to NEDA method shall be given in figure 3.

Use the filter material, making no chemical reaction with the components in flue gas, such as silica wool, or alkaliess glass wool.

- 3) In order to prevent the moisture contained in the sample gas from being condensed, provide a device to heat the place from gas sampling tube to three-way cock (Q).

In addition, to connect these parts, use commonly ground connecting tube, silicone rubber tube or polytetrafluoroethylene resin tube. Set up the line of piping as short as possible, and if it is apprehended that moisture will condense, heat to about 120 °C along the tube from the sampling tube to three-way cock (Q).

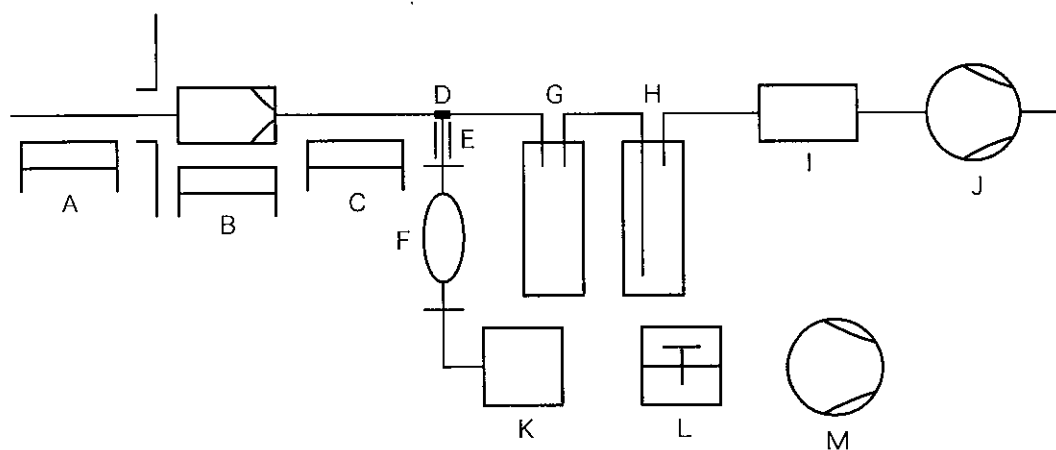
- 4) Set up the whole apparatus so that there is no leakage at any connection.



- | | |
|--|--|
| A : Gas sampling tube | G : Vacant bottle (to prevent back flow) |
| B : Thermal insulating material | H : Washing bottle [containing sodium hydroxide solution (40 g/L)] |
| C : Filter material | I : Drying pump ^{a)} |
| D : Heater | J : Suction pump |
| E : Gas sampling evacuated flask [evacuated flask (E)] | Q, R : Three-way cocks |
| F : Closed tube mercury manometer | S ₁ , S ₂ , S ₃ : Silicone rubber tubes |

Note ^{a)} For desiccant, use granular silica gel or calcium chloride.

a) Zn-NEDA method, ion chromatography, PDS method

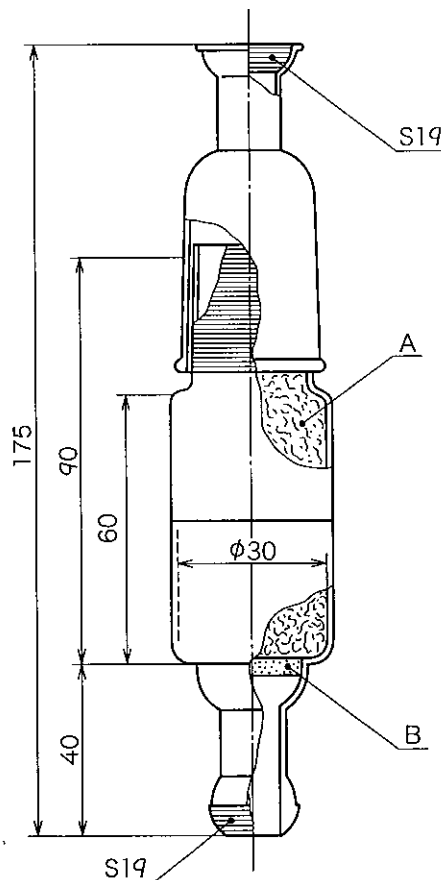


- | | |
|--|--|
| A : Sampling probe | H : Washing bottle [containing sodium hydroxide solution (40 g/L)] |
| B : Filter material | I : Drying tube ^{a)} |
| C : Sampling line | J : Suction pump |
| D : T-piece | K : Pressure meter |
| E : Capillary | L : Thermometer |
| F : Gas sampling flask | M : Vacuum pump |
| G : Vacant bottle (to prevent back-flow) | |

b) NEDA method

Figure 2 Example of sampling apparatus of sample gas

Unit: mm

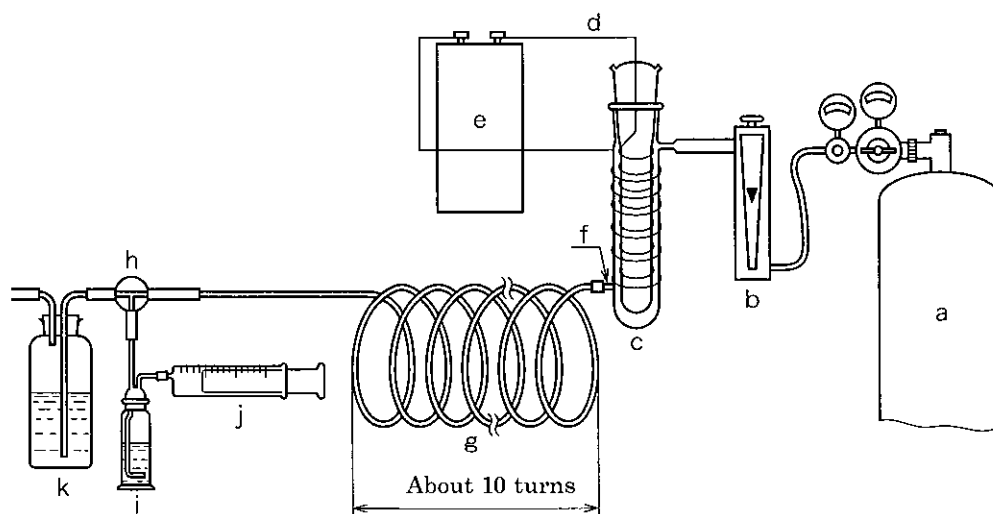


- A : Quartz wool (about 0.5 g to 0.8 g)
 B : Perforated plate or sintered filter
 (capable of supporting quartz wool)

Figure 3 Example of particle filter

- d) **Ozone generator** As illustrated in figure 4, capable of obtaining ozone concentration of about 1 ml /100 ml. It shall have the least production of nitrogen oxides, and the variation of production amount shall be very small.

Upon generating ozone, nitrogen oxides generate at the same time. Since the nitrogen oxides thus generated must be removed by converting fully to dinitrogen pentaoxide (N_2O_5) through the reaction with ozone, they shall be forced to pass through the polytetrafluoroethylene tube (g) of about 10 m in length and furthermore the bottle (i) containing sodium hydroxide solution (40 g/L) to absorb nitrogen oxides. In addition, the sodium hydroxide solution (40 g/L), when showing a high value in the blank test, shall be replaced with the fresh solution.



- | | |
|--|---|
| a : High pressure vessel containing oxygen | g : Ethylene tetrafluoride resin tube for oxidation of nitrogen oxides (for example, a coil of about 10 turns made of a tube of 6 mm in inside diameter, 8 mm in outside diameter and 10 m in length) |
| b : Flow meter | h : Three-way cock |
| c : Ozone generating bottle | i : Bottle for absorbing nitrogen oxides, containing sodium hydroxide solution (40 g/L) |
| d : Copper wire (1 mm in diameter) | j : Ozone adding syringe (M) |
| e : Neon transformer | k : Water seal bottle |
| f : Inlet of ozone-containing oxygen | |

Figure 4 Example of ozone generator

6.1.3 Procedures for sampling

The procedures for sampling shall be carried out as follows.

However, in case of NEDA method, the procedures in a) and b) shall not be carried out. Furthermore, in case of sampling with the evacuated flask with two cocks in accordance with NEDA method, the sampling by using the apparatus constructed as shown in b) of figure 2 shall be performed, but the procedures for sampling shall be carried out in accordance with the method described in the following.

In addition, symbols shown in following shall be in accordance with a) of figure 2.

- a) Remove the lid with cock from the dried evacuated flask (E). inject 20 ml of any absorption solution specified in 4) of b) of 6.1.1 for each analysis method with a syringe (L: see figure 7) or with a pipette, and put the lid with cock on the evacuated flask (E).

The lid and cock of the evacuated flask (E) shall not be greased too much. Fluorocarbon type of grease, etc. which is inactive against nitrogen oxides shall be used.

- b) Connect the evacuated flask (E) to the vacuum pump as shown in figure 2. Open the three-way cock (Q) to the side of the vacuum pump [⊕²], reduce the pressure in the evacuated flask (E) until the absorption solution boils, and then close the three-way cock (Q) (⊕).

Since it does not always boil, shake the flask gently in that case.

Note ²⁾ The directions of the cocks shown hereinafter are based on the arrangement of a) of figure 2.

- c) Turn the three-way cock (R) the side of the closed tube mercury manometer (F) (⊕) [the cock (Q) directions still as shown by ⊕] and suction the sample gas with the suction pump (J) until the moisture in the tube does not condense to water.
- d) Prior to the gas sampling, turn the three-way cock (R) to the side of the evacuated flask (E) (⊕), then, turn the cock (Q) to the side of the closed tube mercury manometer (F)(⊕), and measure and record the pressure (P_1) in the evacuated flask (E). Since, in this case, there is possibility of leakage, if the pressure is 10.1 kPa or over, the ground part shall be regreased, and carry out the procedure after b) again.
- e) Return the three-way cocks (Q) and (R) to their original position (⊖, ⊕). At the same time, measure and record the ambient temperature (t) of the evacuated flask (E).
- f) Turn the three-way cock (Q) to the side of the evacuated flask (E) (⊕) to allow the sample gas to enter the evacuated flask (E), and close the cock (Q) (⊕), and stop the vacuum pump (J).

In addition, for the procedure in case of NEDA method, turn the three-way cock (R) to the side of the evacuated flask (⊕), and then turn the three-way cock (Q) to the side of the evacuated flask and the pressure meter (⊕), take the sample gas, and close the cock when the pressure becomes 10 kPa to 60 kPa according to the concentration of NO_x in the sample gas (⊕).

- g) Take the evacuated flask (E) out of the sampling apparatus of sample gas at the silicone rubber tubes (S_1 , S_2) and shake the flask for 1 min to mix.
- h) Leave the evacuated flask (E) still until it cools down to the room temperature, connect the three-way cock (Q) with the open tube mercury manometer, open the cock (Q) to the connecting side and read the mercury level difference on the manometer, find the pressure (P) in the evacuated flask (E), and then close the cock (Q). At the same time, measure and record the atmospheric pressure (P_a) and the ambient temperature (t) of the room. In case of NEDA method, use the pressure meter in 2) of b) of 6.1.2.

In f) and h), the absolute pressure in the flask is the atmospheric pressure reduced by the mercury level difference. If the pressure difference in the flask is positive, add it to the atmospheric pressure, and subtract if negative.

6.1.4 Preparation of sample solution for analysis

The preparation shall be carried out as follows.

- a) In case of zinc reduction naphthylethylenediamine absorptiometry (Zn-NEDA method)

- 1) As illustrated in figure 4, take 100 ml of ozone-containing oxygen produced by ozone generator into the syringe (M) for adding ozone, immediately connect the syringe (M) to the evacuated flask (E) containing the sample gas taken according to 6.1.3, open the three-way cock (Q) to the side of the syringe (M), and push the inner cylinder of the syringe (M) gently to force ozone-containing oxygen to enter the evacuated flask (E). Close the three-way cock (Q), and then remove the syringe (M).
- 2) Shake the evacuated flask (E) for 1 min to mix, and then leave it still for about 5 min. Furnish the liquid thus obtained to the analysis as the sample solution.

b) In case of naphthylethylenediamine absorptiometry (NEDA method)

- 1) Connect the syringe (L) containing 50 ml of absorption solution and 5 ml of copper sulfate solution which have been previously mixed to the three-way cock (Q) of the evacuated flask (E) with silicone rubber tube, open the three-way clock (Q) to the side of the syringe, put the mixed solution into the evacuated flask ³⁾.

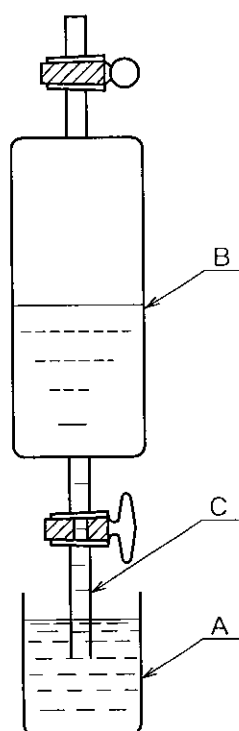
In addition, when using the evacuated flask with two cocks, the absorption solution may be sucked in by the residual pressure inside the evacuated flask as shown in figure 5.

- 2) Close the three-way cock (Q), detach the syringe (L) from the evacuated flask (E), immediately shake the evacuated flask (E) vigorously for 3 min to mix, or after shaking vigorously for 1 min shake it for 2 min using a shaker.

In order to make a vapour phase reaction with NO, NO₂ and SO₂ minimized, carried out procedures from gas sampling to shaking within 5 min.

- 3) Transfer the content solution in the evacuated flask into a 100 ml one-mark volumetric flask, furthermore wash the evacuated flask (E) with 15 ml of water twice and transfer the washing solution also into the flask.
- 4) Open the stopper of the one-mark volumetric flask, and after leaving it still in the water bath at 80 °C for 30 min, take the flask out of the water bath, make it cool with running water down to the room temperature, add water up to the marked line, and then shake it with its stopper closed. Furnish the liquid thus obtained to the analysis as the sample solution.

Note ³⁾ The rubber glove should be used for protecting hands from the alkaline solution.



A : Glass beaker containing absorption solution
B : Flask for gas sampling
C : Silicone rubber tube (approx. 5 cm in length)

Figure 5 Example of injection of absorption solution

c) In case of ion chromatography

1) When oxidizing with ozone

1.1) Take 100 ml of ozone-containing oxygen produced by ozone generator into the syringe (M) for adding ozone, immediately connect the syringe (M) to the evacuated flask (E) containing the sample gas taken according to 6.1.3, open the three-way cock (Q) to the side of the syringe (M), and push the inner cylinder of the syringe (M) gently to force ozone-containing oxygen to enter the evacuated flask (E). Close the three-way cock (Q), and then remove the syringe (M).

1.2) Shake the evacuated flask (E) for 1 min to mix, and then leave it still for about 5 min.

1.3) Transfer the content solution in the evacuated flask (E) to a 100 ml one-mark volumetric flask, add water up to the marked line. Furnish the solution thus obtained to the analysis as the sample solution.

2) **When oxidizing with oxygen** Follow 1). However, for 1.1), use 150 ml of oxygen in place of oxygen containing ozone, and for 1.2), leave still for 2 h or over.

3) **When oxidizing with the residual oxygen in the sample gas** After leaving the flask containing the sample gas taken at 6.1.3 still for 16 h or over, follow 1.2) to 1.3).

d) In case of phenol disulfonic acid absorptiometry (PDS method)

- 1) **When oxidizing with ozone** Follow 1) to 2) of a).
- 2) **When oxidizing with oxygen** Follow 1) to 2) of a). However, use 150 ml of oxygen in place of oxygen containing ozone, and leave the flask still for 2 h or over.
- 3) **When oxidizing with the residual oxygen in the sample gas** After leaving the flask containing the sample gas taken at 6.1.3 still for 16 h or over, shake it for 1 min to mix, and leave it still for about 5 min. Furnish the solution thus obtained to the analysis as the sample solution.

6.1.5 Amount of sample gas taken

According to the following formulae, the amount of sample gas taken under the standard condition [273.15 K (0 °C), 101.32 kPa] shall be converted into dry-base gas amount (V_{SD}) or wet-base gas amount (V_{SW}).

a) In case of obtaining amount of dry-base gas

- 1) **In case of Zn-NEDA method, ion chromatography absorptiometry (Zn-NEDA method), ion chromatography and phenol disulfonic acid absorptiometry (PDS method).**

$$V_{SD} = V_u \times \frac{273.15}{101.32} \times \left(\frac{P_f - P_{wf}}{273.15 + t_f} - \frac{P_i - P_{wi}}{273.15 + t_i} \right) + 22.41(a + b)$$

- 2) **In case of naphthylethylenediamine absorptiometry (NEDA method)**

$$V_{SD} = V_a \times \frac{273.15}{101.32} \times \left(\frac{P_f - P_{wf}}{273.15 + t_f} - \frac{P_i}{273.15 + t_i} \right)$$

b) In case of obtaining amount of wet-base gas

- 1) **In case of Zn-NEDA method, ion chromatography absorptiometry (Zn-NEDA method), ion chromatography and phenol disulfonic acid absorptiometry (PDS method).**

$$V_{SW} = V_u \times \frac{273.15}{101.32} \times \left(\frac{P_f - P_{wf}}{273.15 + t_f} - \frac{P_i - P_{wi}}{273.15 + t_i} \right) + 22.41(a + b + c)$$

- 2) **In case of naphthylethylenediamine absorptiometry (NEDA method)**

$$V_{SW} = V_a \times \frac{273.15}{101.32} \times \left(\frac{P_f - P_{wf}}{273.15 + t_f} - \frac{P_i}{273.15 + t_i} \right) + 22.41c$$

Where,

- V_{SD} : dry-base gas amount (ml)
- V_{SW} : wet-base gas amount (ml)
- V_u : volume value obtained by subtracting quantity of absorption solution (20 ml) from unobstructed capacity of evacuated flask (V_a ml) (ml)
- V_a : unobstructed capacity of evacuated flask (ml)
- P_f : pressure inside evacuated flask after sampling gas and, leaving still (kPa)
- P_{wf} : saturated water vapour pressure at t_f °C (kPa)
- P_i : pressure inside evacuated flask before sampling gas (kPa)
- P_{wi} : saturated water vapour pressure at t_i °C (kPa)
- t : temperature when measuring P (°C)

- t_f : temperature when measuring P_f (°C)
 a^A : target gas for analysis captured in absorption solution (mol)
 b^A : gases, except target gas to analysis, captured in absorption solution (mol)
 c^A : moisture in the flue gas obtained in clause 6 of JIS Z 8808 (mol)
273.15 : absolute temperature equivalent to 0 °C (K)
101.32 : pressure corresponding to 1 atmospheric pressure (kPa)
22.41 : volume of 1 mmol gas at standard condition (ml)

Note 4) In most cases, it may be ignored.

For P_{nf} and P_{ni} , follow figure 2. When the difference between temperatures (t_i and t_f) measured before and after gas sampling is within 2 °C, the amount of sample taken may be obtained by the following formulae by regarding as $P_{ni} = P_{nf}$.

$$V_{SD} = V_u \times \frac{273.15}{273.15 + t_f} \times \frac{P_f - P_i}{101.32} + 22.41(a + b)$$

$$V_{SW} = V_u \times \frac{273.15}{273.15 + t_f} \times \frac{P_f - P_i}{101.32} + 22.41(a + b + c)$$

Table 2 Saturated water vapour pressure

Unit : Pa

$t/^{\circ}\text{C}$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.	611.21	615.67	620.15	624.67	629.21	633.78	638.38	643.01	647.67	652.36
1.	657.08	661.83	666.61	671.42	676.26	681.14	686.04	690.98	695.94	700.94
2.	705.97	711.03	716.13	721.26	726.41	731.61	736.83	742.09	747.38	752.70
3.	758.06	763.45	768.88	774.34	779.83	785.36	790.92	796.52	802.15	807.82
4.	813.52	819.26	825.03	830.84	836.69	842.57	848.49	854.45	860.44	866.47
5.	872.54	878.64	884.79	890.97	897.19	903.44	909.74	916.07	922.45	928.86
6.	935.31	941.80	948.34	954.91	961.52	968.17	974.86	981.60	988.37	995.19
7.	1 002.0	1 008.9	1 015.9	1 022.9	1 029.9	1 037.0	1 044.1	1 051.2	1 058.4	1 065.7
8.	1 072.9	1 080.3	1 087.6	1 095.1	1 102.5	1 110.0	1 117.6	1 125.2	1 132.8	1 140.5
9.	1 148.2	1 156.0	1 163.8	1 171.7	1 179.6	1 187.6	1 195.6	1 203.7	1 211.8	1 219.9
10.	1 228.1	1 236.4	1 244.7	1 253.0	1 261.4	1 269.9	1 278.4	1 286.9	1 295.5	1 304.2
11.	1 312.9	1 321.7	1 330.5	1 339.3	1 348.2	1 357.2	1 366.2	1 375.3	1 384.4	1 393.5
12.	1 402.8	1 412.1	1 421.4	1 430.8	1 440.2	1 449.7	1 459.3	1 468.9	1 478.5	1 488.2
13.	1 498.0	1 507.8	1 517.7	1 527.7	1 537.7	1 547.7	1 557.9	1 568.0	1 578.3	1 588.6
14.	1 598.9	1 609.3	1 619.8	1 630.3	1 640.9	1 651.6	1 662.3	1 673.0	1 683.9	1 694.8
15.	1 705.7	1 716.7	1 727.8	1 739.0	1 750.2	1 761.4	1 772.8	1 784.2	1 795.6	1 807.1
16.	1 818.7	1 830.4	1 842.1	1 853.9	1 865.8	1 877.7	1 889.7	1 901.7	1 913.8	1 926.0
17.	1 938.3	1 950.6	1 963.0	1 975.5	1 988.0	2 000.6	2 013.3	2 026.0	2 038.8	2 051.7
18.	2 064.7	2 077.7	2 090.8	2 104.0	2 117.2	2 130.5	2 143.9	2 157.4	2 170.9	2 184.5
19.	2 198.2	2 212.0	2 225.8	2 239.7	2 253.7	2 267.8	2 281.9	2 296.1	2 310.4	2 324.8
20.	2 339.2	2 353.8	2 368.4	2 383.1	2 397.8	2 412.7	2 427.6	2 442.6	2 457.7	2 472.9
21.	2 488.2	2 503.5	2 518.9	2 534.4	2 550.0	2 565.7	2 581.4	2 597.3	2 613.2	2 629.2
22.	2 645.3	2 661.5	2 677.7	2 694.1	2 710.5	2 727.1	2 743.7	2 760.4	2 777.2	2 794.1
23.	2 811.0	2 828.1	2 845.2	2 862.5	2 879.8	2 897.2	2 914.8	2 932.4	2 950.1	2 967.9
24.	2 985.8	3 003.7	3 021.8	3 040.0	3 058.3	3 076.6	3 095.1	3 113.6	3 132.3	3 151.1
25.	3 169.9	3 188.9	3 207.9	3 227.0	3 246.3	3 265.6	3 285.1	3 304.6	3 324.3	3 344.0
26.	3 363.9	3 383.8	3 403.9	3 424.0	3 444.3	3 464.7	3 485.2	3 505.7	3 526.4	3 547.2
27.	3 568.1	3 589.1	3 610.2	3 631.5	3 652.8	3 674.2	3 695.8	3 717.4	3 739.2	3 761.1
28.	3 783.1	3 805.2	3 827.4	3 849.7	3 872.2	3 894.7	3 917.4	3 940.2	3 963.1	3 986.1
29.	4 009.2	4 032.5	4 055.8	4 079.3	4 102.9	4 126.6	4 150.5	4 174.4	4 198.5	4 222.7
30.	4 247.0	4 271.5	4 296.0	4 320.7	4 345.5	4 370.5	4 395.5	4 420.7	4 446.0	4 471.5
31.	4 497.0	4 522.7	4 548.5	4 574.5	4 600.5	4 626.7	4 653.1	4 679.5	4 706.1	4 732.8
32.	4 759.7	4 786.7	4 813.8	4 841.0	4 868.4	4 895.9	4 923.6	4 951.4	4 979.3	5 007.4
33.	5 035.6	5 063.9	5 092.4	5 121.0	5 149.7	5 178.6	5 207.7	5 236.8	5 266.2	5 295.6
34.	5 325.2	5 355.0	5 384.8	5 414.9	5 445.1	5 475.4	5 505.9	5 536.5	5 567.2	5 598.1
35.	5 629.2	5 660.4	5 691.8	5 723.3	5 754.9	5 786.8	5 818.7	5 850.8	5 883.1	5 915.5
36.	5 948.1	5 980.8	6 013.7	6 046.8	6 080.0	6 113.3	6 146.9	6 180.5	6 214.4	6 248.4
37.	6 282.5	6 316.9	6 351.3	6 386.0	6 420.8	6 455.8	6 490.9	6 526.2	6 561.7	6 597.3
38.	6 633.1	6 669.1	6 705.2	6 741.5	6 778.0	6 814.7	6 851.5	6 888.5	6 925.6	6 963.0
39.	7 000.5	7 038.2	7 076.0	7 114.1	7 152.3	7 190.7	7 229.2	7 268.0	7 306.9	7 346.0
40.	7 385.3	7 424.8	7 464.4	7 504.2	7 544.3	7 584.5	7 624.8	7 665.4	7 706.2	7 747.1

Table 2 (continued)

Unit : Pa

$t/^{\circ}\text{C}$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
41.	7 788.2	7 829.6	7 871.1	7 912.8	7 954.6	7 996.7	8 039.0	8 081.5	8 124.1	8 167.0
42.	8 210.0	8 253.2	8 296.7	8 340.3	8 384.1	8 428.2	8 472.4	8 516.8	8 561.5	8 606.3
43.	8 651.3	8 696.5	8 742.0	8 787.6	8 833.5	8 879.5	8 925.8	8 972.3	9 018.9	9 065.8
44.	9 112.9	9 160.2	9 207.7	9 255.5	9 303.4	9 351.6	9 399.9	9 448.5	9 497.3	9 546.3
45.	9 595.6	9 645.0	9 694.7	9 744.6	9 794.7	9 845.0	9 895.6	9 946.4	9 997.4	10 049.
46.	10 100.	10 152.	10 204.	10 256.	10 308.	10 361.	10 414.	10 467.	10 520.	10 573.
47.	10 627.	10 681.	10 735.	10 790.	10 845.	10 899.	10 955.	11 010.	11 066.	11 122.
48.	11 178.	11 234.	11 291.	11 348.	11 405.	11 462.	11 520.	11 578.	11 636.	11 694.
49.	11 753.	11 812.	11 871.	11 930.	11 990.	12 049.	12 110.	12 170.	12 231.	12 292.
50.	12 353.	12 414.	12 476.	12 538.	12 600.	12 663.	12 725.	12 788.	12 852.	12 915.
51.	12 979.	13 043.	13 107.	13 172.	13 237.	13 302.	13 368.	13 433.	13 499.	13 566.
52.	13 632.	13 699.	13 766.	13 833.	13 901.	13 969.	14 037.	14 106.	14 175.	14 244.
53.	14 313.	14 383.	14 453.	14 523.	14 594.	14 665.	14 736.	14 807.	14 879.	14 951.
54.	15 023.	15 096.	15 169.	15 242.	15 316.	15 389.	15 464.	15 538.	15 613.	15 688.
55.	15 763.	15 839.	15 915.	15 991.	16 068.	16 145.	16 222.	16 299.	16 377.	16 455.
56.	16 534.	16 613.	16 692.	16 771.	16 851.	16 931.	17 012.	17 093.	17 174.	17 255.
57.	17 337.	17 419.	17 501.	17 584.	17 667.	17 750.	17 834.	17 918.	18 003.	18 087.
58.	18 173.	18 258.	18 344.	18 430.	18 516.	18 603.	18 690.	18 778.	18 866.	18 954.
59.	19 043.	19 131.	19 221.	19 310.	19 400.	19 491.	19 581.	19 672.	19 764.	19 856.
60.	19 948.	20 040.	20 133.	20 226.	20 320.	20 414.	20 508.	20 603.	20 698.	20 793.
61.	20 889.	20 985.	21 082.	21 179.	21 276.	21 374.	21 472.	21 571.	21 669.	21 769.
62.	21 868.	21 968.	22 069.	22 170.	22 271.	22 372.	22 474.	22 577.	22 679.	22 783.
63.	22 886.	22 990.	23 094.	23 199.	23 304.	23 410.	23 516.	23 622.	23 729.	23 836.
64.	23 944.	24 052.	24 160.	24 269.	24 379.	24 488.	24 598.	24 709.	24 820.	24 931.
65.	25 043.	25 155.	25 268.	25 381.	25 494.	25 608.	25 723.	25 837.	25 953.	26 068.
66.	26 184.	26 301.	26 418.	26 535.	26 653.	26 772.	26 890.	27 010.	27 129.	27 249.
67.	27 370.	27 491.	27 612.	27 734.	27 857.	27 979.	28 103.	28 226.	28 351.	28 475.
68.	28 600.	28 726.	28 852.	28 979.	29 106.	29 233.	29 361.	29 489.	29 618.	29 748.
69.	29 877.	30 008.	30 138.	30 270.	30 402.	30 534.	30 667.	30 800.	30 933.	31 068.
70.	31 202.	31 338.	31 473.	31 609.	31 746.	31 883.	32 021.	32 159.	32 298.	32 437.
71.	32 577.	32 717.	32 858.	32 999.	33 140.	33 283.	33 425.	33 569.	33 713.	33 857.
72.	34 002.	34 147.	34 293.	34 439.	34 586.	34 734.	34 882.	35 030.	35 179.	35 329.
73.	35 479.	35 630.	35 781.	35 933.	36 085.	36 238.	36 391.	36 545.	36 700.	36 855.
74.	37 010.	37 166.	37 323.	37 480.	37 638.	37 796.	37 955.	38 115.	38 275.	38 436.
75.	38 597.	38 758.	38 921.	39 084.	39 247.	39 411.	39 576.	39 741.	39 907.	40 073.
76.	40 240.	40 408.	40 576.	40 744.	40 914.	41 084.	41 254.	41 425.	41 597.	41 769.
77.	41 942.	42 116.	42 290.	42 464.	42 640.	42 815.	42 992.	43 169.	43 347.	43 525.
78.	43 704.	43 884.	44 064.	44 245.	44 426.	44 608.	44 791.	44 974.	45 158.	45 343.
79.	45 528.	45 714.	45 900.	46 088.	46 275.	46 464.	46 653.	46 843.	47 033.	47 224.
80.	47 416.	47 608.	47 801.	47 994.	48 189.	48 384.	48 579.	48 776.	48 972.	49 170.

Table 2 (concluded)

Unit : Pa

$t/^{\circ}\text{C}$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
81.	49 368.	49 567.	49 767.	49 967.	50 168.	50 370.	50 572.	50 775.	50 979.	51 183.
82.	51 388.	51 594.	51 800.	52 007.	52 215.	52 424.	52 633.	52 843.	53 053.	53 265.
83.	53 477.	53 689.	53 903.	54 117.	54 332.	54 547.	54 764.	54 981.	55 198.	55 417.
84.	55 636.	55 856.	56 076.	56 298.	56 520.	56 743.	56 966.	57 190.	57 415.	57 641.
85.	57 868.	58 095.	58 323.	58 552.	58 781.	59 011.	59 242.	59 474.	59 707.	59 940.
86.	60 174.	60 409.	60 644.	60 881.	61 118.	61 356.	61 594.	61 834.	62 074.	62 315.
87.	62 557.	62 799.	63 042.	63 286.	63 531.	63 777.	64 024.	64 271.	64 519.	64 768.
88.	65 017.	65 268.	65 519.	65 771.	66 024.	66 278.	66 532.	66 788.	67 044.	67 301.
89.	67 559.	67 817.	68 077.	68 337.	68 598.	68 860.	69 123.	69 386.	69 651.	69 916.
90.	70 182.	70 449.	70 717.	70 986.	71 255.	71 526.	71 797.	72 069.	72 342.	72 616.
91.	72 890.	73 166.	73 442.	73 719.	73 998.	74 277.	74 556.	74 837.	75 119.	75 401.
92.	75 685.	75 969.	76 254.	76 540.	76 827.	77 115.	77 404.	77 693.	77 984.	78 276.
93.	78 568.	78 861.	79 155.	79 450.	79 746.	80 043.	80 341.	80 640.	80 940.	81 240.
94.	81 542.	81 844.	82 148.	82 452.	82 757.	83 064.	83 371.	83 679.	83 988.	84 298.
95.	84 609.	84 921.	85 234.	85 547.	85 862.	86 178.	86 495.	86 812.	87 131.	87 451.
96.	87 771.	88 093.	88 415.	88 739.	89 063.	89 389.	89 715.	90 043.	90 371.	90 701.
97.	91 031.	91 362.	91 695.	92 028.	92 363.	92 698.	93 035.	93 372.	93 711.	94 050.
98.	94 391.	94 732.	95 075.	95 418.	95 763.	96 109.	96 455.	96 803.	97 152.	97 502.
99.	97 853.	98 204.	98 557.	98 911.	99 266.	99 623.	99 980.	100 338.	100 697.	101 058.
100.	101 419.	101 782.	102 145.	102 510.	102 875.	103 242.	103 610.	103 979.	104 349.	104 720.

NOTE : Follow SONNTAG (1990). For the scale for temperature, follow ITS-90.

6.2 Syringe method

The syringe method shall be as follows.

This shall be applied to Zn-NEDA method, NEDA method, ion chromatography and PDS method.

6.2.1 Preparation of reagents and reagent solution

Follow 6.1.1.

6.2.2 Implements and apparatus

a) Gas sampling implements

- 1) **Syringe for gas sampling** The glass syringe with glass cock of 500 ml or 200 ml in capacity, one differentiated according to Annex H of JIS K 0050 [hereafter referred to as "syringe (K)"].
- 2) **Syringe for injecting absorption solution** Follow 2) of a) of 6.1.2.
- 3) **Syringe for adding ozone or oxygen** Follow 3) of a) of 6.1.2.

b) Sampling apparatus for sample gas

The apparatus constructed as illustrated in figure 6 for the positions of S_1 to S_3 in ones constructed as illustrated in a) of figure 2.

c) Ozone generator Follow d) of 6.1.2 (figure 4).

6.2.3 Procedures for sampling

The procedures for sampling shall be carried out as follows.

The symbols shown here shall be in accordance with a) of figure 2 and figure 6.

- a) Open the cock (P) of the syringe (K), and suction several ml of the absorption solution. After pulling the inner cylinder slowly with the syringe tip facing upward and with the cock open to the maximum graduation, push the inner cylinder slowly again to moisten fully the inner wall with the absorption solution ⁵⁾. Next, after expelling the absorption solution and air, close the cock (P).

Note ⁵⁾ It will avoid the leakage of gas to moisture the inner wall of the syringe with the absorption solution in advance.

- b) Connect the syringe (K) with silicone rubber tube (S_4) to the apparatus. Open the three-way cock (R) upward ($\ominus$), make the suction pump (J) work, and thereby suction the sample gas until the moisture in the tube does not condense any more.
- c) Close the three-way cock (R) ($\oplus$), immediately open the cock (P) of the syringe (K), take the sample gas by pulling the inner cylinder, and close the cock (P). Return the three-way cock (R) to the original place ($\ominus$), stop the suction pump (J). The sample gas shall amount to about 400 ml when the syringe (K) is 500 ml in capacity and amount to about 150 ml when the syringe (K) is 200 ml in capacity.

Pulling operation of the inner cylinder shall be carried out by one action.

- d) Remove the syringe (K) from the apparatus at the part of the silicone rubber tube (S_4), leave it still until it cools down to the room temperature, and then read the graduation (V_a) of the sample gas taken. At the same time, measure and record the atmospheric pressure (P_a) and the room temperature (t_a).
- e) After injecting 20 ml of the absorption solution of any one of 4.1) to 4.4) of b) of 6.1.1 specified in each analysis method used for the syringe (L), connect to the syringe (K) as shown in figure 7, open the cock (P) of the syringe (K), and push the inner cylinder of the syringe (L) to inject the absorption solution. After closing the cock (P), detach the syringe (L), and immediately shake the syringe (K) for 1 min to mix. However, the quantity of absorption solution in case of NEDA method shall be equal to that in b) of 6.1.1. Carry out procedures from gas sampling, injecting the absorption solution to shaking to mix within 10 min in total.

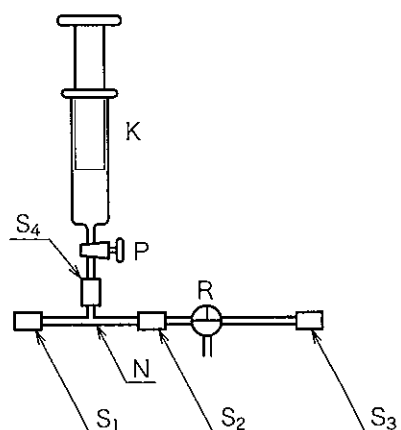


Figure 6 Example of sampling apparatus for sample gas

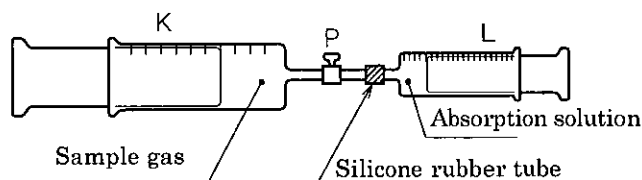


Figure 7 Example of injection of absorption solution

K : Syringe for gas sampling (K)
 P : Cock for syringe (K)
 L : Syringe
 N : T tube
 R : Three-way cock
 S₁, S₂, S₃, S₄ : Silicone rubber tube

6.2.4 Preparation of sample solution for analysis

The preparation shall be carried out as follows.

a) In case of zinc reduction naphthylethylenediamifl absorptiometry (Zn-NEDA method)

- 1) Take 30 ml of ozone-containing oxygen produced by ozone generator into the syringe (M) for adding ozone, immediately connect to the syringe (K) containing the sample gas taken according to 6.2.3, open the cock (P) and push gently the inner cylinder of the syringe (M) to force ozone-containing oxygen to enter the syringe (K). Close the cock (P), and then remove the syringe (M).
- 2) Shake the syringe (K) for 1 min to mix, and then leave it still for about 5 min. Furnish the liquid thus obtained to the analysis as the sample solution.

b) In case of naphthylethylenediamifl absorptiometry (NEDA method)

- 1) Transfer the content solution in the syringe (K) into a 100 ml one-mark volumetric flask, furthermore wash the inside of the syringe with water and transfer washing solution also into the flask.
- 2) Open the stopper of the flask, and after leaving it still in the water bath at 80 °C for 30 min, take the flask out of the water bath, make it cool with running water down to the room temperature, add water up to the marked line, and then shake it well with its stopper closed. Furnish the liquid thus obtained to the analysis as the sample solution.

c) In case of ion chromatography

1) When oxidizing with ozone

- 1.1) Follow the procedures in 1) of a).
- 1.2) Shake the syringe (K) for 1 min to mix, and then leave it still for about 5 min.
- 1.3) Transfer the content solution in the syringe (K) into a 100 ml one-mark volumetric flask, add water up to the marked line. Furnish the solution thus obtained to the analysis as the sample solution.

2) When oxidizing with oxygen Follow the procedures in 1.1) to 1.3). However, for 1.1), use about 20 ml of oxygen for 100 ml of the amount of sample gas taken in place of oxygen containing ozone, and for 1.2), leave the syringe still for 2 h or over.

3) When oxidizing with the residual oxygen in the sample gas After leaving the syringe containing the sample gas taken in 6.2.3 still for 16 h or over, follow the procedures in 1.2) to 1.3).

d) In case of phenol disulfonic acid absorptiometry (PDS method)

1) When oxidizing with ozone Follow the procedures in 1.1) to 1.2) of c).

2) When oxidizing with oxygen Follow the procedures in 1.1) to 1.2) of c). However, for 1.1) of c) use about 20 ml of oxygen for 100 ml of the amount of sample gas taken in place of oxygen containing ozone, and for 1.2) of c), leave the syringe still for 2 h or over.

3) When oxidizing with the residual oxygen in the sample gas After leaving the syringe containing the sample gas taken in 6.2.3 still for 16 h or over, shake it for 1 min to mix, and leave it still for about 5 min. Furnish the solution thus obtained to the analysis as the sample solution.

6.2.5 Amount of sample gas taken

According to the following formulae, the amount of sample gas taken under the standard condition (273.15 K, 101.32 kPa) shall be converted into dry-base gas amount (V_{SD}) or wet-base gas amount (V_{SW}).

a) In case of obtaining amount of dry-base gas

$$V_{SD} = V_a \times \frac{273.15}{273.15 + t_s} \times \frac{P_a - P_{as}}{101.32}$$

b) In case of obtaining amount of wet-base gas

$$V_{SW} = V_a \times \frac{273.15}{273.15 + t_s} \times \frac{P_a - P_{as}}{101.32} + 22.41c$$

where,	V_{SD} :	dry-base gas amount (ml)
	V_{SW} :	wet-base gas amount (ml)
	V_a :	amount of the sample gas sucked in the syringe (K) (ml)
	P_a :	atmospheric pressure (kPa)
	P_{ns} :	saturated water vapour pressure at t_s °C (kPa)
	t_s :	room temperature after sampling gas and leaving still (°C)
	c :	moisture obtained in clause 6 of JIS Z 8808 (mol)

6.3 Absorption bottle method

This method shall be applied to Saltzman absorptiometry.

6.3.1 Preparation of reagents and reagent solution

a) Reagents

- 1) Acetic acid Specified in JIS K 8355.
- 2) Sulfanilic acid Specified in JIS K 8586.
- 3) *N*-1-naphthylethylenediamine dihydrochloride Specified in JIS K 8197.
- 4) Nitrogen Grade 2 nitrogen specified in JIS K 1107.

b) Preparation of reagent solution

- 1) *N*-1-naphthylethylenediamine dihydrochloride solution (1 g/L) Dissolve 0.1 g of *N*-1-naphthylethylenediamine dihydrochloride in 100 ml of water.
- 2) Absorption colouring liquid Dissolve 5 g of sulfanilic acid in about 900 ml of warm water containing 50 ml of acetic acid. After cooling the solution, add 50 ml of *N*-1-naphthylethylenediamine dihydrochloride solution (1 g/L), add water to make 1 000 ml in total, and preserve the solution thus obtained in a brown bottle.

6.3.2 Implements and apparatus

a) Gas sampling implements

- 1) Absorption bottle The bottle with the glass ball filter attached to the tip of capillary tube. The example is shown in figure 8.
- 2) Syringe for gas sampling The syringe of 200 ml in capacity specified in JIS T 3201, one previously differentiated according to Annex H of JIS K 0050 [hereafter referred to as "syringe (F)"].
- 3) Syringe for adding nitrogen The syringe of 200 ml in capacity specified in JIS T 3201.

b) Sampling apparatus for sample gas The apparatus, constructed as shown in figure 9, shall be equipped with the conditions which are equivalent to the provisions of c) of 6.1.2.

6.3.3 Procedures for sampling

The procedures for sampling shall be as follows.

The symbols shown here shall, be in accordance with figure 9.

- Take 25 ml of the absorption colouring liquid, prepared in 2) of b) of 6.3.1 into an absorption bottle.
- Close the stop cock (J), make the suction pump (I) work, and replace the inside of tube with the sample gas sufficiently.
- Connect the absorption bottle (E) and the syringe (F) to the sampling apparatus for sample gas.
- Open the stop cock (J), suck about 100 ml of the sample gas by using the syringe (F).

Take 5 ml of the solution for working curve for nitrogen dioxide into a 25 ml one-mark volumetric flask, and add the absorption. Colouring liquid up to the marked line and keep the liquid at hand. If the sample gas colours darker than this liquid, reduce the amount of sample gas taken at the next time.

- Close the stop cock (J), and read the graduation of sample gas taken (V_a). At the same time, record, the atmospheric pressure (P_a) and the ambient temperature (t).
- Detach the absorption bottle (E) and the syringe (F) from the sampling apparatus for sample gas.

Unit : mm

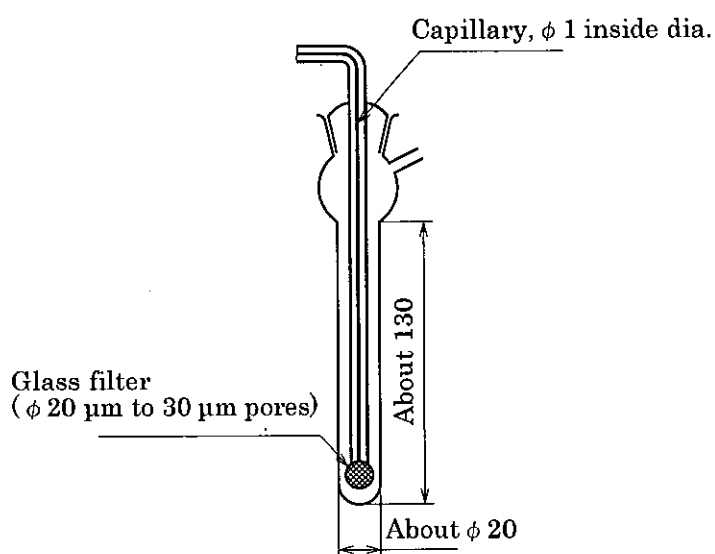
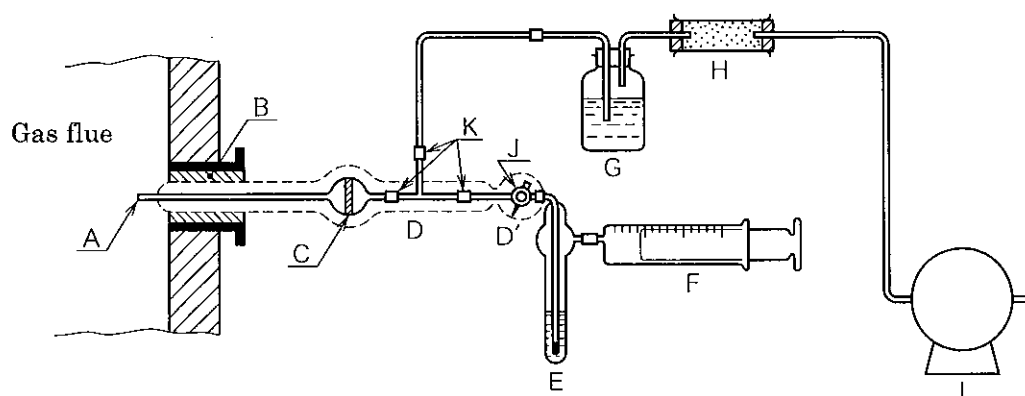


Figure 8 Example of absorption bottle



- | | |
|---------------------------------|---------------------------------------|
| A : Gas sampling tube | F : Gas sampling syringe |
| B : Thermal insulating material | G : Washing bottle (containing water) |
| C : Filter material | H : Drying tube |
| D : Heater | I : Suction pump |
| D' : Heater | J : Stop cock |
| E : Absorption bottle | K : Silicone rubber tube |

Figure 9 Example of sampling apparatus for gas sample

6.3.4 Preparation of sample solution for analysis

The preparation shall be carried out as follows.

- a) Take 200 ml of nitrogen into the syringe for adding nitrogen.
- b) Connect the syringe containing nitrogen to the inlet side of absorption bottle, force nitrogen to pass through at a flow rate of about 100 ml /min and expel the residual nitrogen monoxide in the inside space of the absorption bottle. Furnish the liquid thus obtained to the analysis as the sample solution.

Carry out the expelling procedure of nitrogen monoxide immediately after sampling gas. Furthermore, since the sample solution for analysis prepared here is already coloured, carry out the measurement of absorbance according to 7.5.3 after 20 min later.

6.3.5 Amount of sampling gas taken

Follow 6.2.5.

7 Chemical analysis method

7.1 Zinc reduction naphthylethylenediamine absorptiometry (Zn-NEDA method)

7.1.1 Preparation of reagents and reagent solution

a) Reagents, material

- 1) Hydrochloric acid Specified in JIS K 8180.
- 2) Acetic acid Specified in JIS K 8355.
- 3) Sodium hydroxide Specified in JIS K 8576.

- 4) **Sodium acetate trihydrate** Specified in JIS K 8371.
 - 5) **Sulfanilamide** Specified in JIS K 9066.
 - 6) ***N*-1-naphthylethylenediamine dihydrochloride** Specified in JIS K 8197.
 - 7) **Potassium nitrate** Specified in JIS K 8548.
 - 8) **Nitrate-ion standard solution (NO_3^- : 1 000 mg/L)**
 - 9) **Zinc powder** Specified in JIS K 8013, used for the analysis of nitrogen oxides, 90 % or over in reduction rate from nitrate ion to nitrite ion. Zinc powder, in which nitrogen gas is enclosed, is stable in the long term unless the container is opened. If the container is once opened, however, the reduction ratio of zinc powder lowers gradually, and care shall be taken for the protection from oxidation, for example, by sealing nitrogen gas.
 - 10) **Filter paper** Grade 5B specified in JIS P 3801.
- b) **Preparation of reagent solution**
- 1) **Hydrochloric acid (1+1)**
 - 2) **Sodium hydroxide (100 g/L)**
 - 3) **Sulfanilamide solution** Weigh 3.33 g of sulfanilamide in a beaker, add 10 ml of hydrochloric acid (1+1) and 50 ml of water to dissolve it, and add the solution obtained by dissolving 250 g of trihydrate of sodium acetate in about 400 ml of water. Furthermore add acetic acid or sodium hydroxide solution (100 g/L) and after adjusting its pH value to 7 ± 0.1 , add water to make 1 000 ml in total. This solution can be used until it turns to faint yellow.
 - 4) ***N*-1-naphthylethylenediamine dihydrochloride (1 g/L)** Follow 1) of b) of 6.3.1.
 - 5) **Nitrogen oxides standard stock solution (NO_2 : 100 $\mu\text{L/ml}$)** After drying potassium nitrate in the dryer at $105^\circ\text{C} \pm 2^\circ\text{C}$ for 2 h, cool it in a desiccator. Then, take 0.451 g of potassium nitrate, dissolve it in water, transfer it into a 1 000 ml one-mark volumetric flask while rinsing, and add water up to the marked line. The value corresponds to that containing 100 μl of NO_2 with 273.15 K and 101.32 kPa in 1 ml of solution.
 - 6) **Nitrogen oxides standard solution (NO_2 : 1 $\mu\text{l/ml}$)** Take 10 ml of the nitrogen oxides standard stock solution (NO_2 : 100 $\mu\text{l/ml}$) of 5) into a 1 000 ml one-mark volumetric flask, and add water up to the marked line. In place of this solution, nitrate-ion standard solution (NO_3^- : 1 000 mg/L) of 8) of a) may be used by diluting it.

This nitrogen oxides standard solution is prepared when it is needed.

7.1.2 Procedures for determination

The procedures shall be as follows.

- a) Transfer the sample solution for analysis prepared in accordance with a) of 6.1.4 or a) of 6.2.4 a) to a 100 ml one-mark volumetric flask, and add 15 ml of sulfanilamide solution to mix. Add 0.5 g of zinc powder, immediately add water up to the marked line, and then shake it for 1 min to mix them.

Since the shaking condition affects the determined value according to the nature of zinc powder, the shaking condition shall always be constant. It is recommended to obtain the best condition in advance.

- b) Filter the zinc powder which has not reacted through dry filter paper, and after discarding the first filtrate of about 20 ml, receive the remaining filtrate in a dry 100 ml beaker.
- c) Take 20 ml of the filtrate in a 25 ml one-mark volumetric flask, and add 3 ml of hydrochloric acid (1+1) and 1 ml of *N*-1-naphthylethylenediamine dihydrochloride solution, add water up to the marked line, and then shake them to mix well. Leave this solution at 15 °C to 30 °C for about 15 min. If it is anticipated for the sample solution to exceed the range of determination take 10 ml or 5 ml.
- d) Take a part of this solution in an absorption cell, and measure the absorbance in the vicinity of wavelength of 545 nm by taking the blank test solution prepared in accordance with 7.1.3 as the control.
- e) On the working curve prepared in 7.1.4, find the volume (μl) of nitrogen oxides in the sample solution for analysis used in this procedure.

7.1.3 Preparation of blank test solution

- a) **In case of evacuated flask method** Carry out the procedures of a) to h) of 6.1.3 and a) of 6.1.4 using air in place of the sample gas. Successively, carry out the procedures according to a) to c) of 7.1.2.
- b) **In case of syringe method** Carry out the procedures of a) to e) of 6.2.3 and 1) to 2) of a) of 6.2.4 using almost the same amount of air as that of sample gas taken in place of the sample gas. Successively, carry out the procedures according to a) to c) of 7.1.2.

7.1.4 Preparation of working curve

- a) Take stepwise several amounts from 1 ml to 50 ml of the nitrogen oxides standard solution (NO₂ : 1 μl/ml) into as many 100 ml one-mark volumetric flasks, and add 20 ml of absorption solution prepared in 4.1) of b) of 6.1.1 into each flask.
- b) Thereafter, follow the procedure (prescribed in a) to c) of 7.1.2 for the sample prepared in a) and then obtain the absorbance in the vicinity of wavelength of 545 nm by taking the solution prepared without addition of the nitrogen oxides standard solution as the control. Make a curve showing the relation between the volume (μl) of nitrogen oxides and the absorbance thereof.

In the procedure in c) of 7.1.2, 20 ml of filtrate is used.

7.1.5 Calculation

Calculate the concentration of nitrogen oxides in the sample gas according to the following formulae:

$$C_v = \frac{V \times n \times 1\,000}{V_s}$$

$$C_w = \frac{2.05 \times V \times n \times 1\,000}{V_s}$$

$$C_w = C_v \times 2.05$$

where, C_v : volumetric concentration of nitrogen oxides in the sample gas (vol ppm)
 C_w : mass concentration of nitrogen oxides in the sample gas when they are converted into nitrogen dioxide (mg/m³)
 V : volume of nitrogen oxides obtained on the working curve (μl)
 V_s : quantity of sample gas taken calculated according to 6.1.5 or 6.2.5 (ml)
 (V_{SD} in case of dry-base gas amount, and V_{SW} in case of wet-base gas amount)
 n : solution correction [this value n shall be 1 for 20 ml of filtrate in c) of 7.1.2, 2 for 10 ml, 4 for 5 ml.]
 2.05: mass concentration of nitrogen dioxide (NO₂) equivalent to 1 vol ppm of nitrogen oxides (mg/m³), (46.01/22.41)

7.2 Naphthylethylenediamine absorptiometry (NEDA method)

7.2.1 Preparation of reagents and reagent solution

a) Reagents

- 1) Sulfanilamide Specified in JIS K 9066.
- 2) Hydrochloric acid Specified in JIS K 8180.
- 3) *N*-1-naphthylethylenediamine dihydrochloride Specified in JIS K 8197.
- 4) Sodium nitrite Specified in JIS K 8019.
- 5) Nitrite-ion standard solution (NO₂⁻: 1 000 mg/L)
- 6) Sodium hydroxide Specified in JIS K 8576.

b) Preparation of reagent solution

- 1) Acidic sulfanilamide hydrochloric acid solution (5 g/L) Take 50 ml of water into a 200 ml one-mark volumetric flask, and dissolve 1.0 g of sulfanilamide. Furthermore, after adding 112 ml of hydrochloric acid, add water up to the marked line.
- 2) *N*-1-naphthylethylenediamine dihydrochloride solution (1 g/L) Follow 1) of b) of 6.3.1.
- 3) Nitrite-ion standard stock solution (NO₂⁻: 250 mg/L) ⁶⁾ Heat sodium nitrite at 105 °C ± 2 °C for about 4 h and cool it in desiccator. Obtain the pure grade of sodium nitrite according to JIS K 8019, take 375 mg of sodium nitrite for 100 % of NaNO₂ to dissolve it in a little water. Then add 0.2 g of sodium hydroxide to it, transfer it into a 1 000 ml one-mark volumetric flask with water, and water up to

the marked line.

Note ⁶⁾ If this solution is stored in a bottle stoppered tightly at a cold dark place, it is stable at least for 3 months.

- 4) **Nitrite-ion standard solution (NO_2^- : 20 mg/L)** Take 40 ml of nitrite-ion standard stock solution (NO_2^- : 250 mg/L) of 3) into a 500 ml one-mark volumetric flask, and add water up to the marked line. Otherwise take 2 ml of nitrite-ion standard solution (NO_2^- : 1 000 mg/L) of 5) of a) into a 100 ml one-mark volumetric flask, and add water up to the marked line.

This Nitrite-ion standard solution is prepared when it is needed.

- 5) **Nitrite-ion standard solution (NO_2^- : 100 mg/L)** Take 40 ml of nitrite-ion standard stock solution (NO_2^- : 250 mg/L) of 3) into a 100 ml one-mark volumetric flask, and add water up to the marked line. Otherwise take 10 ml of nitrite-ion standard solution (NO_2^- : 1 000 mg/L) of 5) of a) into a 100 ml one-mark volumetric flask, and add water up to the marked line.

This Nitrite-ion standard solution is prepared when it is needed.

7.2.2 Procedures for determination

Procedures shall be as follows.

- a) Transfer 20 ml of the sample solution prepared in accordance with b) of 6.1.4 or b) of 6.2.4 into a 100 ml one-mark volumetric flask, add 10 ml of acidic sulfanilamide hydrochloric acid solution (5 g/L) and 5 ml of *N*-1-naphthylethylenediamine dihydrochloride solution (1 g/L), and add water up to the marked line. Then shake them well to mix.

Since it is neutralized by adding acidic sulfanilamide hydrochloric acid solution (5 g/L) and carbon dioxide is suddenly generated, add this solution little by little as shaking it in the vicinity of the neutral point.

- b) After leaving this solution at 15 °C to 30 °C for about 15 min, measure the absorbance in the vicinity of wavelength of 545 nm using the solution which has been prepared from the blank test solution prepared in 7.2.3 in the same procedure as the control.

7.2.3 Preparation of blank test solution

Take 50 ml of the absorption solution of 4.2) of b) of 6.1.1 and 5 ml of copper sulfate (II) solution of 2) of b) of 6.1.1 into a 100 ml one-mark volumetric flask. Successively, carry out the procedure in 4) of b) of 6.1.4. The liquid thus obtained shall be used as the blank test solution.

7.2.4 Preparation of working curve

Preparation of working curve shall be as follows.

- a) Take stepwise several amounts from 1 ml to 5 ml of nitrite-ion standard solution (NO_2^- : 20 mg/L) of 4) of b) of 7.2.1 or nitrite-ion standard solution (NO_2^- : 100 mg/L) of 5) of b) of 7.2.1 into respective 100 ml one-mark volumetric flasks according to the anticipated concentration of nitrogen oxide in the sample.

- b) Take 50 ml of absorption solution of 4.2) of b) of 6.1.1 and 5 ml of copper (II) sulfate solution of 2) of b) of 6.1.1 which have been previously mixed into each evacuated flask (E) using the syringe for injecting absorption solution, and shake it vigorously for 3 min, or after shaking vigorously for 1 min, shake it for 2 min using a shaker. In case of the syringe method, use the syringe (K) in place of evacuated flask (E), put the same amount of air as that of sample gas taken.
- c) Transfer this solution into each one-mark volumetric flask of a), and furthermore, wash each evacuated flask (E) with 15 ml of water twice, transfer the washing solution also into the flask.
- d) Open the stopper of the flask, and after leaving it still in the water bath at 80 °C for 30 min, take the flask out of the water bath, cool it down to the room temperature in running water, and then add water up to the marked line, shake it well with its stopper closed. The one-mark volumetric flask used for the procedure should not be used for the other applications.
- e) Hereinafter, carry out the procedures in accordance with 7.2.2.

NOTE : Sufficient care shall be taken for the following interferences.

- a) Since sulfur dioxide (SO_2) reacts with carbon monoxide (NO), carbon dioxide (NO_2) and becomes interfering substance in a gas phase, prevent it from interfering by transferring it into a liquid phase to make it be sulfate ion (SO_4^{2-}).
- b) If the concentration of hydrochloric acid in the flue gas is not more than 365 mg/m³, it has no effect on analysis.
- c) The existence of copper (II) ion (Cu^{2+}) in high concentration suppresses the generation of azo dye.
- d) CO , CO_2 are not interfering substance.
- e) Dinitrogen monoxide (N_2O) is not interfering substance in the range within the concentration of 100 mg/m³ (No investigation in the range of the higher concentration has been carried out.)
- f) Ammonia (NH_3) is not interfering substance in the range within the concentration of 20 mg/m³ (No investigation in the range of the higher concentration has been carried out.)
- g) Sulfur trioxide (SO_3) is not interfering substance in the range within the concentration of 100 mg/m³, hydrogen fluoride (HF) within the concentration of 10 mg/m³ (No investigation in the range of the higher concentration has been carried out.)

7.2.5 Calculation

Calculate the concentration of nitrogen oxides in the sample gas according to the following formulae:

$$C_v = \frac{0.487 \times V}{V_s} \times \frac{100}{V'} \times 10^6$$

$$C_w = \frac{V}{V_s} \times \frac{100}{V'} \times 10^6$$

- where,
- C_v : volumetric concentration of nitrogen oxides in the sample gas (vol ppm)
 - C_w : mass concentration of nitrogen oxides in the sample gas (mg/m³)
 - V : mass of nitrogen dioxide obtained on the working curve (mg)
 - V' : amount of sample solution for analysis taken (20 ml)
 - V_s : amount of sample gas taken (ml)
(V_{sd} in case of dry-base gas amount, and V_{sw} in case of wet-base gas amount)
 - 0.487 : volume of nitrogen dioxide equivalent to 1 mg of nitrogen dioxide (ml)

The lower limit and repeatability (standard deviation under repeatable conditions %) in this analysis method shall be as follows.

- Lower limit of determination: 2 mg/m³ in case of using the cell of 5 cm in optical path length
- Repeatability (standard deviation under repeatable conditions %): within 3 % in the level of concentration of NO_x of about 50 mg/m³ to 1 000 mg/m³

7.3 Ion chromatography

7.3.1 Preparation of reagents and reagent solution

a) Reagents

- 1) Sulfuric acid Specified in JIS K 8951.
- 2) Boric acid Specified in JIS K 8868.
- 3) Sodium hydrogen carbonate Specified in JIS K 8622.
- 4) Sodium carbonate Specified in JIS K 8625.
- 5) Sodium tetraborate decahydrate Specified in JIS K 8866.
- 6) Potassium nitrate Specified in JIS K 8548.
- 7) Sodium nitrite Specified in JIS K 8019.
- 8) Plithalic acid
- 9) Potassium gluconate
- 10) Acetonitrile Specified in JIS K 8032.
- 11) Glycerol Specified in JIS K 8295.
- 12) para-hydroxybenzoic acid
- 13) 2-[bis (2-hydroxyethyl) amino]-2-hydroxymethyl-1,3-propanediol (bis-tris) Specified in JIS K 9808.

14) 2-Amino-2-hydroxymethyl-1,3-propanediol Specified in JIS K 9704.

15) Nitrate-ion standard solution (NO_3^- : 1 000 mg/L)

16) Nitrite-ion standard solution (NO_2^- : 1 000 mg/L)

b) Preparation of reagent solution

- 1) **Eluent** Depending on the type of apparatus and separation column, which are to be employed, the eluent each of whose ion to be measured can be separated at 1.3 or over of the degree of separation (R) shall be used. The confirmation of the separation degree shall be as specified in clause 10 of JIS K 0127.

NOTE : Examples of eluents are as follows.

When it is confirmed that nitrate ion and nitrite ion are quantitatively measured with 1.3 or over of the degree of separation depending on the characteristics of separation column, the eluent, which is other than the mentioned here, may be used.

- **Hydrogen carbonate salt — carbonate salt solution (1)** Dissolve 0.025 g (0.3 mmol) of sodium hydrogen carbonate and 0.286 g (2.7 mmol) of sodium carbonate in water, transfer it into a 1 000 ml one-mark volumetric flask while rinsing with water, and add water up to the marked line.
- **Hydrogen carbonate salt — carbonate salt solution (2)** Dissolve 0.143 g (1.7 mmol) of sodium hydrogen carbonate and 0.191 g (1.8 mmol) of sodium carbonate in water, transfer it into a 1 000 ml one-mark volumetric flask while rinsing with water, and add water up to the marked line.
- **Gluconate salt — tetraborate salt — boric acid solution** Dissolve 0.305 g (1.3 mmol) of potassium gluconate, 0.496 g (1.3 mmol) of sodium tetraborate decahydrate, 1.855 g (30 mmol) of boric acid, 100 ml of acetonitrile, and 5 ml of glycerol in water, transfer it into a 1 000 ml one-mark volumetric flask while rinsing with water, and add water up to the marked line.
- ***Para*-hydroxybenzoic acid-bis — 2-[bis (2-hydroxyethyl) amino]-2-hydroxymethyl-1,3-propanediol solution** Dissolve 1.105 g (8.0 mmol) of *para*-hydroxybenzoic acid and 0.669 g (3.2 mmol) of 2-[bis (2-hydroxyethyl) amino]-2-hydroxymethyl-1,3-propanediol solution (bis-tris) in water, transfer it into a 1 000 ml one-mark volumetric flask while rinsing with water, and add water up to the marked line.
- **Phthalic acid — aminohydroxymethylpropanediol solution** Dissolve either 0.415 g (2.5 mmol) of phthalic acid and 0.290 g (2.4 mmol) of 2-amino-2-hydroxymethyl-1, 3-propanediol, or 0.382 g (2.3 mmol) of phthalic acid and 0.303 g (2.5 mmol) of 2-amino-2-hydroxymethyl-1, 3-propanediol in water, transfer it into a 1 000 ml one-mark volumetric flask while rinsing with water, and add water up to the marked line.

- 2) **Regenerant (remover)** The solution to be used for recover and continuing maintenance of the function of suppressors. It is employed when the function is recovered electrically or chemically and the most suitable one shall be used for apparatus and suppressors.

NOTE : Examples of regenerant and recycled materials are as follows.

- a) **Water** Decompose by an electric current the specified water in e) of clause 4, and use it for a suppressor which is capable of generating regenerant.
 - b) **Eluent** Decompose by an electric current the eluent passing through a detector, and use it for a suppressor which is capable of producing regenerant.
 - c) **Sulfuric acid (12.5 mmol/L)** Dilute 12.5 ml of sulfuric acid (1 mol/L) with water to make total 1 L. This solution is used as the regenerant.
 - d) **Ion-exchange resin** Mix a cationic exchanger with the eluent.
- 3) **Nitrate-ion standard stock solution (NO_3^- : 1 mg/ml)** Prepared from nitrate-ion standard solution (NO_3^- : 1 000 mg/ml) in 15) of a) of 7.3.1. Otherwise heat and dry potassium nitrate previously at $105 \pm 2^\circ\text{C}$ for about 4 h, allow it to cool in a desiccator. Weigh its 1.631 g, dissolve it in water, transfer it into a 1 000 ml one-mark volumetric flask and add water up to the marked line. Prepare this solution when it is needed.
- 4) **Nitrate-ion standard solution (NO_3^- : 0.1 mg/ml)** Take accurately 10 ml of nitrate-ion standard stock solution (NO_3^- : 1 mg/ml), prepared at 3), into a 100 ml one-mark volumetric flask, and add water up to the marked line.

Prepare this nitrate-ion standard solution when it is needed.

- 5) **Nitrite-ion standard stock solution (NO_2^- : 1 mg/ml)** Prepared from nitrite-ion standard solution (NO_2^- : 1 000 mg/ml) in 16) of a) of 7.3.1. Otherwise heat and dry sodium nitrite previously at $105 \pm 2^\circ\text{C}$ for about 4 h, allow it to cool in a desiccator. Obtain the pure grade of the nitrite-ion standard stock solution in accordance with JIS K 8019, weigh the sodium nitrate equivalent to 1.500 g against 100 % of NaNO_2 , dissolve it in water, transfer it into a 1 000 ml one-mark volumetric flask and add water up to the marked line.

Prepare this nitrate-ion standard stock solution when it is needed.

- 6) **Nitrite-ion standard solution (NO_2^- : 0.1 mg/ml)** Take 10 ml of nitrite-ion, standard stock solution (NO_2^- : 1 mg/ml), prepared in 5), into a 100 ml one-mark volumetric flask, and add water up to the marked line.

Prepare this nitrate-ion standard solution when it is needed. When it is confirmed that nitrite-ion does not coexist in the sample solution for analysis (the peak of nitrite-ion is not generated in the procedure of 7.3.3), none of the preparation and procedure of reagents related to the measurement of nitrite-ion are required.

7.3.2 Implements and apparatus

- a) **Sample introducing device** Automatic type by which specified quantity of sample solution for analysis is introduced into the apparatus with superior reproducibility, or manual type by which sample solution is introduced into a sample measuring tube (specified quantity of about 10 μ l to 250 μ l), which has been installed on the apparatus, using a syringe with a capacity from 1 ml to 10 ml.
- b) **Ion chromatograph** The ion chromatographs are classed into two types, that the type equipped with both separation column and suppressor, and the type with only separation column without suppressor. Either may be serviceable in this Standard.
 - 1) **Separation column** The column, measuring 2 mm to 8 mm in inside diameter and 30 mm to 300 mm in length, shall be made of inactive synthetic resin or metal, and packed with anion exchanger. It shall be capable of separating the target ion from adjacently existing ion at 1.3 or over of the degree of separation (R).
 - 2) **Pre-column** The column, which is for concentrating, pre-separating, and for alien-matter eliminating, shall be set before the separation column if it is needed. It, measuring 2 mm to 6 mm in inside diameter and 5 mm to 50 mm in length, shall be made of inactive synthetic resin or metal, and packed with the same anion exchanger as that in separation column.
 - 3) **Suppressor** The implement for changing an eluting solvent electrically or chemically and reducing its conductivity, in order to use a conductivity meter and measure the ionic species in an eluting solvent with high sensitivity. A suppressor is available in a film dialysis type, a column removal type, and suspension resin adsorption type.
 - 4) **Detector** Electric-conductivity detector
 - 5) **Recording part** It is as specified in f) of 4.2 of JIS K 0127.

7.3.3 Procedures for determination

The procedures shall be as follows.

- a) Set an ion chromatograph to be ready for measurement, and run the eluent through a separation column at the constant rate (for instance, 1 ml/min to 2 ml/min). When adopting the apparatus with suppressor, run the eluent through the inside of the separation column and suppressor, and furthermore run constantly the regenerant through the outside of the suppressor.
- b) Using a sample introducing device, introduce specified quantity (10 μ l to 250 μ l) of sample solution, for analysis prepared at c) of 6.2.4 into the ion chromatograph, and record its chromatogram.

When especially low concentrated sample is to be measured, instead of the sample measuring tube in the apparatus, the concentrating column should preferably be used which has been packed with anion exchanger of the same type as that in separation column.

- c) Find the peak, corresponding to nitrate ion or nitrite ion, on the chromatogram, and obtain the peak area or the peak height.

- d) On the working curve prepared in 7.3.5, obtain the concentration of nitrate ion (mg/ml) and nitrite ion (mg/ml).
- e) Using 20 ml of absorption solution prepared in 4.3) of b) of 6.1.1, take the solution prepared in 7.3.4 in a 100 ml one-mark volumetric flask, add water up to the marked line, take the same amount as that in b), carry out the procedures according to a) to c), and obtain the blank-test value on nitrate ion or nitrite ion.

In c), d) and e), when it is confirmed that nitrite-ion does not coexist in the sample solution for analysis (the peak of nitrite-ion is not generated in the procedure of 7.3.3), none of the preparation and procedure of reagents related to the measurement of nitrite-ion are required.

7.3.4 Preparation of blank test solution

- a) **In case of evacuated flask method** Carry out the procedures of 6.1.3 and c) of 6.1.4 using air in place of the sample gas. Successively, carry out the procedures according to a) to d) of 7.3.3.
- b) **In case of syringe method** Carry out the procedures of 6.2.3 and c) of 6.2.4 using the almost same amount of air as that of sample gas taken in place of the sample gas. Successively, carry out the procedures according to a) to d) of 7.3.3.

In a) and b), if air contained in flasks and syringes is contaminated by nitrogen oxides, ammonia, etc., use nitrogen gas in place of air.

7.3.5 Preparation of working curve

The preparation of working curve shall be as follows.

a) Nitrate-ion

- 1) Take stepwise several amounts from 0.1 ml to 40 ml of nitrate-ion standard solution (NO_3^- : 0.1 mg/ml) into respective 100 ml one-mark volumetric flasks, add water up to the marked line, and obtain each of their concentration. For stepwise taking, take several amounts either from 0.1 ml to 2.0 ml or from 2.0 ml to 40 ml depending on the anticipated concentration of nitrate-ion in the sample.
- 2) Carry out the procedures from a) to c) of 7.3.3, and find the peak area or peak height which is corresponding to each concentration of nitrate ion.
- 3) As a blank test, separately carry out on water the procedures from b) to c) of 7.3.3, find the peak area or peak height which is corresponding to each nitrate ion to obtain the concentration of nitrate ion (mg/ml). Take the concentration of nitrate ion (mg/ml) when introduced into ion chromatography.
- 4) Prepare the relation curve plotting the relation between the concentration of nitrate ion and the peak area or peak height which has been corrected by the blank-test value.

Prepare the working curve every time when sample is measured.

b) Nitrite ion

- 1) When the peak of nitrite ion is generated in c) of 7.3.3, take stepwise several amounts from 0.05 ml to 20 ml of nitrite-ion standard solution (NO_2^- : 0.1 mg/ml) into as many 100 ml one-mark volumetric flasks, add water up to the marked line, and obtain each of their concentration.

For stepwise taking, take several amounts either from 0.05 ml to 1.0 ml or from 1.0 ml to 20 ml depending on the anticipated concentration of nitrite ion in the sample.

- 2) Carry out the procedures from a) to c) of 7.3.3, and find each peak area or peak height which is respectively corresponding to each concentration of nitrite ion.
- 3) As a blank test, separately carry out on water the procedures from b) to c) of 7.3.3, find the peak area or peak height which is corresponding to each nitrite ion to obtain the concentration of nitrite ion (mg/ml).

When it is confirmed that nitrite-ion does not coexist in the sample solution for analysis (the peak of nitrite-ion is not generated in the procedure of 7.3.3), none of the preparation and procedure of reagents related to the measurement of nitrite-ion are required.

- 4) Prepare the relation curve plotting the relation between the concentration of nitrate ion and the peak area or peak height which has been corrected by the blank-test value.

Prepare the working curve every time when sample is measured.

7.3.6 Calculation

Calculate the concentration of nitrogen oxides in the sample gas according to the following formulae:

$$C_v = \frac{[0.361 \times (a_1 - b_1) + 0.487 \times (a_2 - b_2)] \times 100}{V_s} \times 10^6$$

$$C_w = \frac{[0.742 \times (a_1 - b_1) + 1.000 \times (a_2 - b_2)] \times 100}{V_s} \times 10^6$$

$$C_w = C_v \times 2.05$$

- where,
- C_v : volume fraction of nitrogen oxides in the sample gas (vol ppm)
 - C_w : mass concentration of nitrogen oxides in the sample gas when they are converted into nitrogen dioxide (mg/m³)
 - a_1 : concentration of nitrate ion in the sample solution for analysis obtained in d) of 7.3.3 (mg/ml)
 - b_1 : concentration of nitrate ion in blank test obtained in e) of 7.3.3 (mg/ml)
 - a_2 : concentration of nitrite ion in the sample solution for analysis obtained in d) of 7.3.3 (mg/ml)
 - b_2 : concentration of nitrite ion in blank test obtained in e) of 7.3.3 (mg/ml)

V_s :	quantity of sample gas calculated according to 6.1.5 or 6.2.5 (ml) (V_{SD} in case of dry-base gas amount, and V_{SW} in case of wet-base gas amount)
0.361 :	volume of nitrogen oxides equivalent to 1 mg of nitrate ion (NO_3^-) (ml) (standard condition)
0.487 :	volume of nitrogen oxides equivalent to 1 mg of nitrite ion (NO_2^-) (ml) (standard condition)
0.742 :	mass of nitrogen dioxide (NO_2) equivalent to 1 mg of nitrate ion (NO_3^-) (mg)
1.000 :	mass of nitrogen dioxide equivalent to 1 mg of nitrite ion (NO_2^-) (mg)
2.05 :	mass concentration of nitrogen dioxide (NO_2) equivalent to 1 vol ppm of nitrogen oxides (mg/m^3), (46.01/22.41)

7.4 Phenol disulfonic acid absorptiometry (PDS method)

7.4.1 Applicable condition

This method is affected by the high concentrated coexistence of halogen compounds in the sample gas, therefore this can be applicable when such influence can be neglected or eliminated.

7.4.2 Preparation of reagents, materials and reagent solution

a) Reagents, material

- 1) Sulfuric acid Specified in JIS K 8951.
- 2) Sulfuric acid, fuming Sulfuric acid having 15 % to 30 % of free SO_3 specified in JIS K 8741.
- 3) Ammonia solution Specified in JIS K 8085.
- 4) Sodium hydroxide Specified in JIS K 8576.
- 5) Potassium hydroxide Specified in JIS K 8574.
- 6) Phenol Specified in JIS K 8798.
- 7) Nitrate-ion standard solution (NO_3^- :1 000 mg/L)
- 8) Filter paper Grade B of type 5 specified in JIS P 3801.

b) Preparation of reagent solution

- 1) Sodium hydroxide solution (250 g/L)
- 2) Potassium hydroxide solution (56 g/L)
- 3) Phenol disulfonic acid solution Add 25 g of phenol in a beaker containing 150 ml of sulfuric acid, heat the beaker on a water bath to melt phenol. After cooling, add 75 ml of fuming sulfuric acid with care, heat the liquid on a water bath for 2 h,

and after cooling preserve the liquid in a brown bottle.

- 4) **Nitrogen oxides standard solution (NO_2 : 10 $\mu\text{l/ml}$)** Take 10 ml of nitrogen oxides standard stock solution (NO_2 : 100 $\mu\text{l/ml}$) of 5) of b) of 7.1.1 into a 100 ml one-mark volumetric flask, and add water up to the marked line. In place of this solution, nitrate-ion standard solution (NO_3^- : 1 000 mg/L) of 7) of a) may be used by diluting it.

Prepare this nitrate oxides standard solution when it is needed.

7.4.3 Procedures for determination

Procedures shall be carried out as follows.

- a) Transfer the sample solution for analysis prepared in accordance with d) of 6.1.4 or d) of 6.2.4 into an evaporating dish while rinsing.

If it is anticipated for the sample solution to exceed the range of determination, take quantity of 1/2 or 1/4 of the sample solution for analysis partly, and add 10 ml or 15 ml of the absorption solution prepared in 4.4) of b) of 6.1.1 respectively.

- b) Add the potassium hydroxide solution (56 g/L) to the solution in the evaporating dish until it becomes alkaline based on a litmus paper test. Evaporate to dryness on the water bath, and after potassium sulfate is separated as crystals, furthermore heat it on the water bath for about 10 min.
- c) After cooling it, add 2 ml of phenol disulfonic acid solution little by little carefully, and contact the residue with the solution well by crushing it finely with a glass rod. Add 1 ml of water and 4 drops of sulfuric acid, and heat it in the water bath for 3 min stirring frequently.
- d) After cooling it, add 10 ml of water, and, stir it sufficiently. Furthermore, drop 15 ml of cold ammonia water or sodium hydroxide solution (250 g/L).
- e) Immediately filter the solution using a filter paper, and receive the filtrate in a brown 100 ml one-mark volumetric flask. Wash the evaporating dish with a little water, and filter while washing the filter paper with this washing solution. Repeat this procedure several times, and add water up to the marked line.
- f) Take a part of this solution in an absorption cell, and measure the absorbance in the vicinity of wavelength of 400 nm by taking the blank test solution prepared in accordance with 7.4.4 as the control.
- g) On the working curve prepared in 7.4.5, obtain the volume (μl) of nitrogen oxides in the sample solution for analysis used in this procedure.

7.4.4 Preparation of blank test solution

- a) **In case of evacuated flask method** Carry out the procedures in a) to h) of 6.1.3 and d) of 6.1.4 using air in place of the sample gas. Successively, carry out the procedures according to a) to e) of 7.4.3.

- b) **In case of syringe method** Carry out the procedures in a) to e) of 6.2.3 and d) of 6.2.4 using almost the same amount of air as that of sample gas taken in place of the sample gas. Successively, carry out the procedures according to a) to e) of 7.4.3.

7.4.5 Preparation of working curve

- a) Take stepwise several amounts from 0 ml to 30 ml of nitrogen oxides standard solution (NO_2 : 10 $\mu\text{l/ml}$) into respective evaporating dishes, add 20 ml of the absorption solution prepared in 4.4) of b) of 6.1.1 to each dish.
- b) Thereafter, follow the procedure prescribed in b) to e) of 7.4.3 using the sample solution prepared in a) and then obtain the absorbance in the vicinity of wavelength of 400 nm by taking the solution prepared without addition of the nitrogen oxides standard solution as the control. Make a curve showing the relation between the volume (μl) of nitrogen oxides and the absorbance thereof.

7.4.6 Calculation

Calculate the concentration of nitrogen oxides in the sample gas according to the following formulae:

$$C_v = \frac{V \times n \times 1000}{V_s}$$

$$C_w = \frac{2.05 \times V \times n \times 1000}{V_s}$$

- where,
- C_v : volumetric concentration of nitrogen oxides in the sample gas (vol ppm)
 - C_w : mass concentration of nitrogen oxides in the sample gas when they are converted into nitrogen dioxide (mg/m^3)
 - V : volume of nitrogen oxides obtained on the working curve (μl)
 - V_s : quantity of sample gas at standard condition calculated according to 6.1.5 or 6.2.5 (ml)
(V_{SD} in case of dry-base gas amount, and V_{SW} in case of wet-base gas amount)
 - n : solution correction [this value n shall be 1 for total quantity of the sample solution for analysis taken in a) of 7.4.3, 2 for quantity of 1/2, 4 for quantity of 1/4].
 - 2.05 : mass concentration of nitrogen dioxide (NO_2) equivalent to 1 vol ppm of nitrogen oxides (mg/m^3) (46.01/22.41)

7.5 Saltzman absorptiometry

7.5.1 Applicable condition

This method is affected by the high concentrated coexistence of nitrogen monoxide in the sample gas, therefore this can be applicable when such influence can be neglected or eliminated.

7.5.2 Preparation of reagents and reagent solution

a) Reagents

- 1) **Sodium nitrite** Specified in JIS K 8019.

2) Nitrite-ion standard solution (NO_2^- :1 000 mg/L)

b) Preparation of reagent solution

Nitrogen dioxide standard solution (NO_2 :4 $\mu\text{l}/\text{ml}$) Heat and dry sodium nitrite at about 105 ± 2 °C for about 4 h in advance, and allow it to cool in a desiccator. Obtain the pure grade of the sodium nitrite in accordance with JIS K 8019, weigh the sodium nitrite equivalent to 0.259 g against 100 % of NaNO_2 , dissolve in water, transfer it into a 1 000 ml one-mark volumetric flask, and add water up to the marked line. Take 20 ml of this solution into a 500 ml one-mark volumetric flask, and add water up to the marked line. In place of this solution, nitrite-ion standard solution (NO_2^- :1 000 mg/L) in 2) of a) may be used by diluting it.

7.5.3 Procedures for determination

Procedures shall be as follows.

- After leaving the sample solution for analysis in 6.3.4 for 20 min from sampling, measure the absorbance in the vicinity of wavelength of 545 nm using the absorption colouring liquid prepared in 2) of b) of 6.3.1 as the control.
- On the working curve prepared in 7.5.4, obtain the volume (μl) of nitrogen dioxide in the sample solution for analysis.

7.5.4 Preparation of working curve

- Take stepwise several amounts from 0 ml to 5 ml of nitrogen dioxide standard solution (NO_2 :4 $\mu\text{l}/\text{ml}$) into respective 25 ml one-mark volumetric flasks, add the absorption colouring liquid to each flask and leave them for 20 min.
- Obtain the absorbance in the vicinity of wavelength of 545 nm by taking the absorption colouring liquid as the control. Make a curve showing the relation between the volume (μl) of nitrogen dioxide and the absorbance thereof.

7.5.5 Calculation

The concentration of nitrogen dioxide in the sample gas shall be calculated according to the following formulae:

$$C_v = \frac{V}{V_s} \times 1\,000$$

$$C_w = \frac{2.05 \times V}{V_s} \times 1\,000$$

where, C_v : volumetric concentration of nitrogen dioxide in the sample gas (vol ppm)
 C_w : mass concentration, of nitrogen dioxide in the sample gas (mg/m^3)
 V : volume of nitrogen dioxide in the sample solution for analysis obtained on the working curve (μl)
 V_s : quantity of sample gas at standard condition calculated according to 6.3.5 (ml)
 (V_{SD} in case of dry-base gas amount, and V_{SW} in case of wet-base gas amount)

2.05 : mass concentration of nitrogen dioxide (NO₂) equivalent to 1 vol ppm of nitrogen dioxide (mg/m³),
(46.01/22.41)

In this method, the Saltzman factor shall be 0.84.

8 Automated measuring method

Automated measuring method shall be carried out by using measuring instruments specified in JIS B 7982. Calibration shall be carried out in accordance with JIS K 0055.

9 Recording of analysis results

9.1 Items to be recorded

Items to be recorded as analysis results shall be as follows.

- a) Analysis value
- b) Type of analysis method
- c) Date and hour of sampling gas
- d) Other items thought to be necessary

9.2 Method to obtain analysis value

9.2.1 In case of chemical analysis

Carry out twice determination on the same sample solution prepared by every sampling, calculate their average values, and round it off to two significant figures. However, when determining by using the total quantity of the sample solution for analysis, round the analysis value measured at every sampling off to two significant figures.

9.2.2 In case of automated measuring method

Make the average of values measured for 1 h, read the value to the nearest 1/100 of the largest graduation on measurable range, and round it off to two significant figures. When adopting digital indication or print recording, read the figure which corresponds to it.

Annex JA (normative)

Simultaneous analysis of nitrogen oxides, sulfur oxides and hydrogen chloride by ion chromatography

JA.1 Scope

This Annex specifies the simultaneous analysis method of nitrogen oxides, sulfur oxides and hydrogen chloride by ion chromatography in 7.3.

JA.2 Outline of analysis method

Outline of analysis method is as given in table JA.1.

Table JA.1 Summary of analysis method

Summary	Sampling method	Determination range vol ppm(mg/m ³)		
		Nitrogen oxides	Sulfur oxides	Hydrogen chloride
After making nitrogen oxides, sulfur oxides and hydrogen chloride in the sample gas absorb into absorption solution, introduce it into an ion chromatograph, and record its chromatogram.	Evacuated flask method	4 to 1 400 (8 to 2 800)	5 to 1 100 (14 to 3 100)	4 to 630 (6 to 1 000)
	Syringe method	19 to 7 200 (39 to 14 000)	24 to 5 800 (69 to 16 000)	16 to 3 100 (26 to 5 000)

JA.3 Gas sampling method

JA.3.1 Preparation of reagents and reagent solution

- Hydrogen peroxide Follow 6) of a) of 6.1.1.
- Absorption solution

Hydrogen peroxide (1 + 99) solution Add 990 ml water to 10 ml hydrogen peroxide. Store it in dark and cold place.

JA.3.2 Implements and apparatus

Follow 6.1.2 or 6.2.2.

JA.3.3 Sampling

Follow 6.1.3 or 6.2.3.

JA.3.4 Preparation of sample solution for analysis

Collect simultaneously the nitrogen oxides, sulfur oxides and hydrogen chloride existing in flue gas into the absorption solution of b) of JA.3.1, and prepare the solution according to the method in c) of 6.1.4 or c) of 6.2.4.

JA.3.5 Calculation of amount of sample gas taken

Follow 6.1.5 or 6.2.5.

JA.4 Analysis method

JA.4.1 Preparation of reagents and reagent solution

- a) **Eluent** Follow 1) of b) of 7.3.1.
- b) **Regenerant** Follow 2) of b) of 7.3.1.
- c) **Sulfate-ion standard solution (SO_4^{2-} :1 000 mg/L)** Class SO_4^{2-} :1 000 traceable to the National Measurement Standard
- d) **Chloride-ion standard solution (Cl^- :1 000 mg/L)** Class Cl^- :1 000 traceable to the National Measurement Standard
- e) **Sulfate-ion standard stock solution (SO_4^{2-} :1 mg/ml)** Prepared from sulfate-ion standard solution (SO_4^{2-} :1 000 mg/L) in c). Otherwise heat previously sodium sulfate at about 110 °C for about 2 h, and allow it to cool in a desiccator. Weigh its 1.479 g, dissolve in water, transfer it into a 1 000 ml one-mark volumetric flask while rinsing with water, and add water up to the marked line. In place of sodium sulfate, potassium sulfate may be used. In this case, heat strong previously at about 750 °C for about 15 min, weigh its 1.814 g after cooling it, carry out by similar method as the above, and make total 1 000 ml.
- f) **Sulfate-ion standard solution (SO_4^{2-} :0.2 mg/ml)** Take 50 ml of sulfate-ion standard stock solution (SO_4^{2-} :1 mg/ml) prepared in e) into a 250 ml one-mark volumetric flask, and add water up to the marked line. Prepare this solution when it is used.
- g) **Chloride-ion standard stock solution (Cl^- :1 mg/ml)** Prepared from chloride-ion standard solution (Cl^- :1 000 mg/l) in d). Otherwise, heat previously sodium chloride at about 600 °C for about 1 h, and allow it to cool in a desiccator. Weigh its 1.648 g, dissolve it in a small amount of water, transfer it into a 1 000 ml one-mark volumetric flask while rinsing with water, and add water up to the marked line.
- h) **Chloride-ion standard solution (Cl^- :0.1 mg/ml)** Take 10 ml of chloride-ion standard stock solution (Cl^- :1mg/ml) into a 100 ml one-mark volumetric flask, and add water up to the marked line. Prepare this solution when it is needed.
- i) **Nitrate ion, sulfate ion and chloride ion mixed standard solution (NO_3^- :0.04 mg/ml, SO_4^{2-} :0.05 mg/ml, Cl^- :0.01 mg/ml)** Take 40 ml of nitrate-ion standard solution (NO_3^- :0.1 mg/ml) of 4) of b) of 7.3.1, 25 ml of sulfate-ion standard solution (SO_4^{2-} :0.2 mg/ml) of f) and 10 ml of chloride-ion standard solution (Cl^- :0.1 mg/ml) of h) into a 100 ml one-mark volumetric flask, and add water up to the marked line. Prepare this solution when it is needed and dilute as appropriate for use.

JA.4.2 Implements and apparatus

Follow 7.3.2.

JA.4.3 Procedures for determination

The procedures shall be as follows.

- a) Carry out procedures according to a) to b) of 7.3.3.
- b) Find the peaks corresponding to nitrate ion, sulfate ion and chloride ion on chromatogram and obtain respective peak areas or peak heights.
- c) Carry out procedures according to e) of 7.3.3, and find blank-test value of nitrate ion, sulfate ion and chloride ion in absorption solution.
- d) On the working curve, find the concentration of nitrate ion, sulfate ion and chloride ion, and calculate the concentration (mg/ml) of nitrate ion, sulfate ion and chloride ion in the sample solution for analysis.

JA.4.4 Preparation of blank test solution

Follow 7.3.4.

JA.4.5 Preparation of working curve

Take nitrate ion, sulfate ion and chloride ion mixed solution prepared at i) of JA.4.1, and prepare the working curve according to 7.3.5.

JA.4.6 Calculation

Calculate the concentration of nitrogen oxides, sulfur oxides and hydrogen chloride in the sample gas as nitrogen dioxide (NO₂), sulfur dioxide (SO₂) and hydrogen chloride (HCl) respectively, according to the following formulae.

- a) The concentration of nitrogen oxides in the sample gas is calculated by the formulae in 7.3.6.
- b) The concentration of sulfur oxides and hydrogen chloride in the sample gas is calculated according to the following formulae:

1) Concentration of sulfur oxide

$$C_v = \frac{0.233 \times (a - b) \times 100}{V_s} \times 10^6$$

$$C_w = \frac{0.667 \times (a - b) \times 100}{V_s} \times 10^6$$

$$C_w = C_v \times 2.86$$

where, C_v : volumetric concentration of sulfur oxides in the sample gas (vol ppm)
 C_w : mass concentration of sulfur oxides in the sample gas when they are converted into sulfur dioxide (mg/m³)
 a : concentration of sulfate ion obtained in procedures for determination of JA.4.3 (mg/ml)
 b : concentration of sulfate ion obtained in procedures for determination of JA.4.4 (mg/ml)
 V_s : quantity of sample gas taken calculated according to JA.3.5 (ml)
 (V_{sd} in case of dry-base gas amount, and V_{sw} in case of wet-base gas amount)
 0.233 : volume of sulfur oxides (SO₂ + SO₃) equivalent to 1 mg of sulfate ion (SO₄²⁻) (ml) (standard condition)

- 0.667 : mass of sulfur dioxide (SO₂) equivalent to 1 mg of sulfate ion (SO₄²⁻)(mg)
 2.86 : mass concentration of sulfur dioxide (SO₂) equivalent to 1 vol ppm of sulfur oxides (mg/m³), (64.06/22.41)

2) Concentration of hydrogen chloride

$$C_v = \frac{0.632 \times (a - b) \times 100}{V_s} \times 10^6$$

$$C_w = \frac{1.03 \times (a - b) \times 100}{V_s} \times 10^6$$

$$C_w = C_v \times 1.63$$

- where,
- C_v : volumetric concentration of hydrogen chloride in the sample gas (vol ppm)
 - C_w : mass concentration of hydrogen chloride in the sample gas (mg/m³)
 - a : concentration of chloride ion obtained in procedures for determination of JA.4.3 (mg/ml)
 - b : concentration of chloride ion obtained in blank test of JA.4.4 (mg/ml)
 - V_s : quantity of sample gas taken calculated according to JA.3.5 (ml) (standard condition)
 (V_{SD} in case of dry-base gas amount, and V_{SW} in case of wet-base gas amount)
 - 0.632 : volume of hydrogen chloride (HCl) equivalent to 1 mg of chloride ion (Cl⁻) (ml) (standard condition)
 - 1.03 : mass of hydrogen chloride (HCl) equivalent to 1 mg of chloride ion (Cl⁻) (mg)
 - 1.63 : mass concentration of hydrogen chloride (HCl) equivalent to 1 vol ppm of hydrogen chloride (mg/m³), (36.46/22.41)

Annex JB (informative)
Comparison table between JIS and corresponding International Standard

JIS K 0104 :2011 Methods for determination of nitrogen oxides in flue gas				ISO 11564:1998 Stationary source emissions—Determination of the mass concentration of nitrogen oxides—Naphthylethylenediamine photometric method			
(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standard		(IV) Classification and details of technical deviation between JIS and the International standard by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content		No. of clause	Content	Classification by clause	Detail of technical deviation	
1 Scope	Specifies the methods to determine nitrogen oxides in flue gas.		1	Specifies a chemical analysis (NEDA method) to determine nitrogen oxides in flue gas.	Addition	Adds JIS-specific methods, such as an ion chromatography used for actual measurement in Japan.	The methods added by JIS will be studied in the drafting process of international standard.
4 General matters	Specifies chemical analysis, flue gas sampling method, absorptiometry, and ion chromatography, etc.		—	—	Addition	Adds reagents and the general matters of JIS-specific methods.	The methods added by JIS will be studied in the drafting process of international standard.
5 Type and summary of analysis methods	Chemical analysis : NEDA method, ion chromatography, etc. Automated measuring method : Chemiluminescence method, infrared-ray absorbing method, etc.		2	Principle of NEDA method	Addition	Adds JIS-specific methods, such as an ion chromatography used for actual measurement in Japan.	The actual measurement methods in Japan are not included in the international standard. So, the methods added by JIS will be studied in the drafting process of international standard.

(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standard		(IV) Classification and details of technical deviation between JIS and the International standard by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content		No. of clause	Content	Classification by clause	Detail of technical deviation	
6 Method of gas sampling	6.1 Evacuated flask method		4 5	Almost same as JIS. However, adds ion chromatography and PDS method.	Alteration Addition	It specifies reagents. Furthermore, it adds JIS-specific methods used for actual measurement in Japan.	The methods added by JIS will be studied in the drafting process of international standard.
	6.2 Syringe method						
	6.3 Absorption bottle method						
7 Chemical analysis method	7.1 Zn-NEDA method		6 7	Almost same as JIS. However, ion chromatography and PDS method are not specified.	Addition	Adds JIS-specific methods being used for actual measurement in Japan.	The actual measurement methods in Japan are not included in the international standard. So, the methods added by JIS will be studied in the drafting process of international standard.
	7.2 NEDA method						
	7.3 Ion chromatography						
	7.4 PDS method						
	7.5 Saltzman absorptiometry						
			—	Same as JIS.	Identical		
			—	—	Addition	Adds JIS-specific method being used for actual measurement in Japan.	The actual measurement method in Japan are not included in the international standard. So, the method added by JIS will be studied in the drafting process of international standard.

(I) Requirements in JIS		(II) International Standard number	(III) Requirements in International Standard		(IV) Classification and details of technical deviation between JIS and the International standard by clause		(V) Justification for the technical deviation and future measures
No. and title of clause	Content		No. of clause	Content	Classification by clause	Detail of technical deviation	
8 Automated measuring method	Specifies the method by using automated measuring instruments of nitrogen compound in flue gas specified in JIS B 7982.		—	—	Addition	Adds JIS-specific methods being used for actual measurement in Japan.	The actual measurement methods in Japan are not included in the international standard. So, the methods added by JIS will be studied in the drafting process of international standard.
9 Recording of analysis results	Specifies items to be recorded.		10	Almost same as JIS.	Addition	Adds the item in the case of automated measurement.	The actual measurement methods in Japan are not included in the international standard. So, the methods added by JIS will be studied in the drafting process of international standard.
Annex JA (normative)	Specifies simultaneous analysis by ion chromatography.		—	—	Addition	Adds JIS-specific methods being used for actual measurement in Japan.	The actual measurement methods in Japan are not included in the international standard. So, the methods added by JIS will be studied in the drafting process of international standard.

Overall degree of correspondence between JIS and International Standards (ISO 11564 : 1998) : MOD	
NOTE 1	Symbols in sub-columns of classification by clause in the above table indicate as follows: — Identical: Identical in technical contents. — Addition: Adds the specification item(s) or content(s) which are not included in International Standards. — Alteration: Alters the specification content(s) which are included in International Standards.
NOTE 2	Symbol in column of overall degree of correspondence between JIS and International Standards in the above table indicates as follows: — MOD: Modifies International Standards.

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